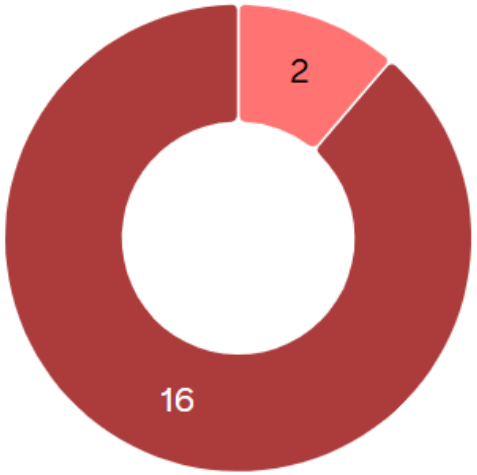
Economics of education

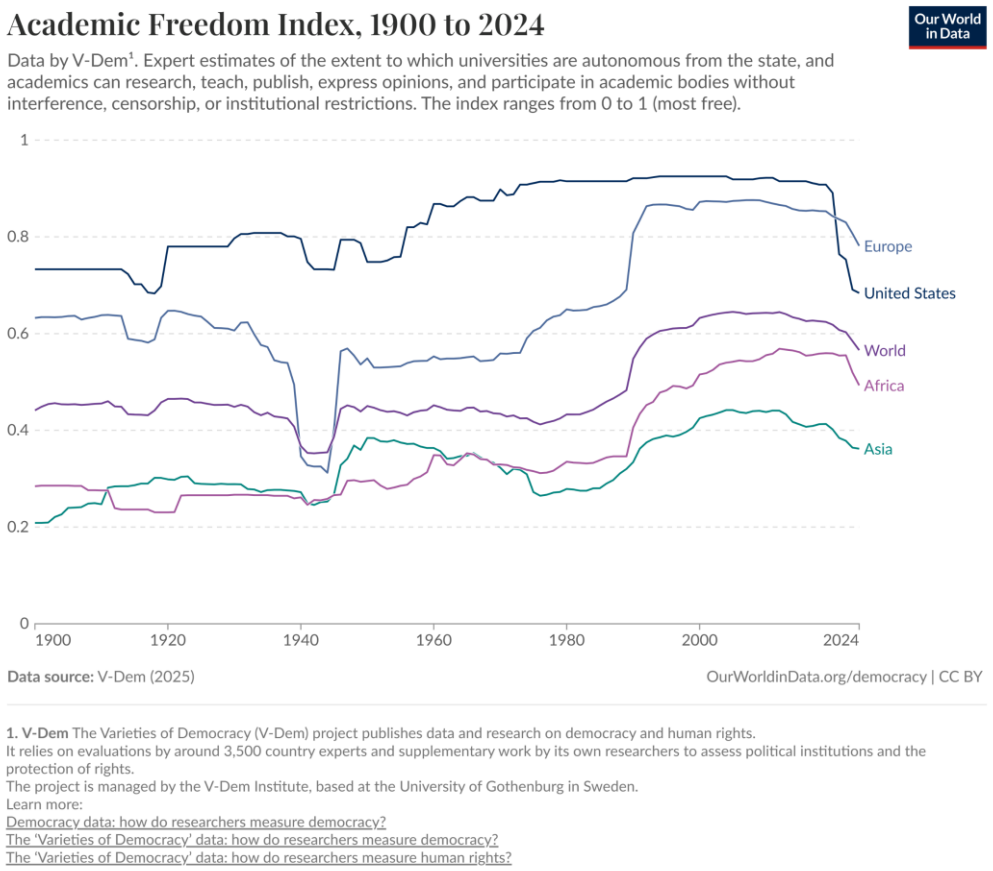
5th class

About last class

What could explain this break in trend according to the model?



0	Education becomes less productive
2	Propaganda technology becomes better
0	Protests become less effective
0	Rule of law worsens
16	All of the above



About last class

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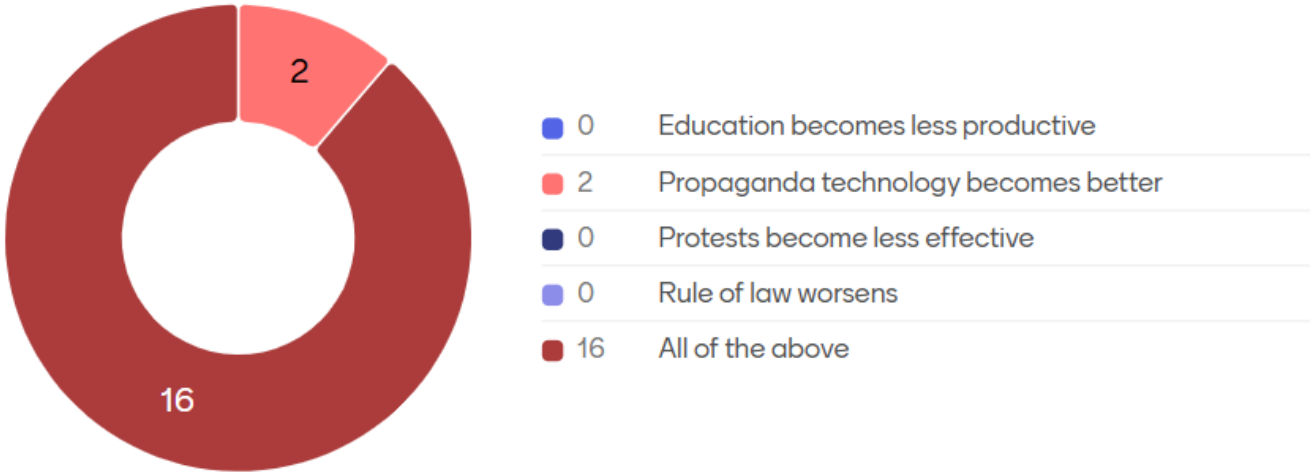

Academic freedom in this context is a proxy for curriculum propaganda. There are restrictions on what researchers can teach, research, and publish. So, less academic freedom, more propaganda.

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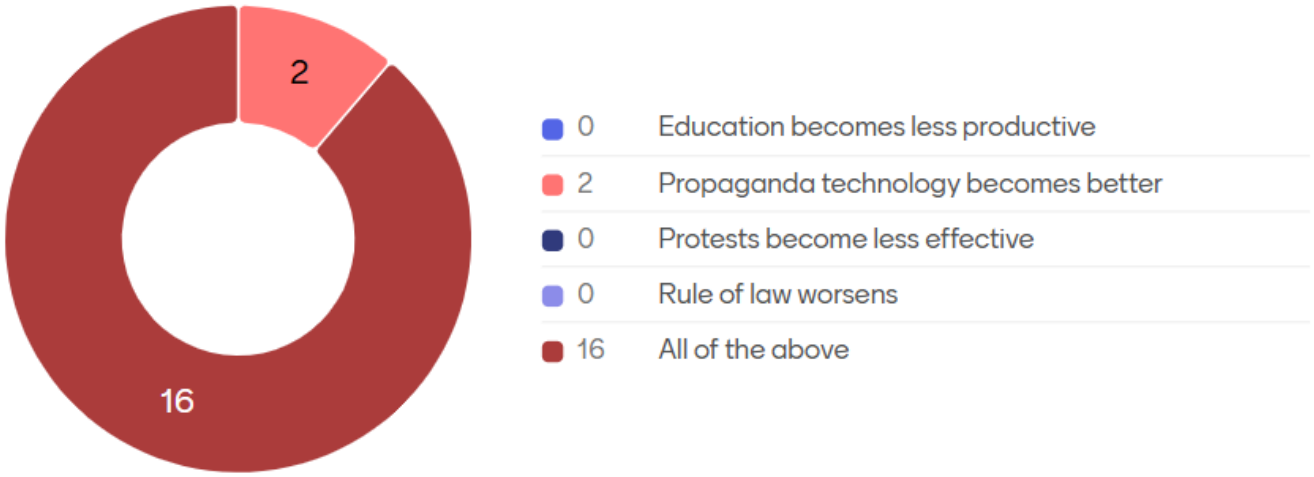
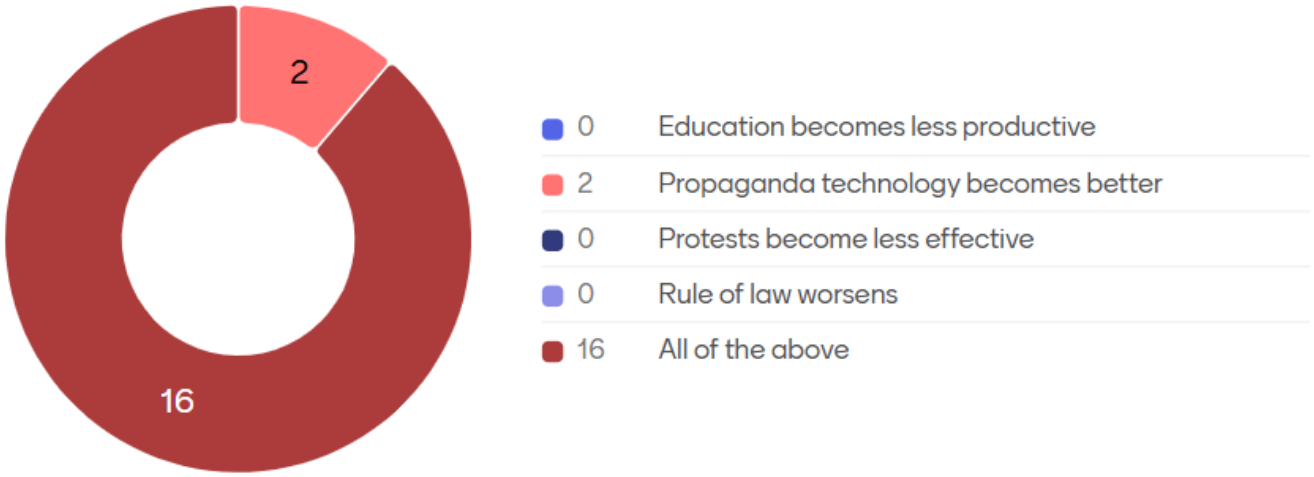


- $t_{e=0}$ = w
- $t_{c=0}$ = w + αh
- $t_{c=1}$ = (1- qσ)(w + αh – αc)

We have seen last class that there are three potential payoffs for the predatory ruler. Increases in propaganda should increase payoff number 2 in comparison to payoff number 3.

About last class

What could explain this break in trend according to the model?



- $t_{e=0}$ = w
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Education becoming less productive increases decreases payoff #2 more than it decreases payoff #3 -> wrong choice

About last class

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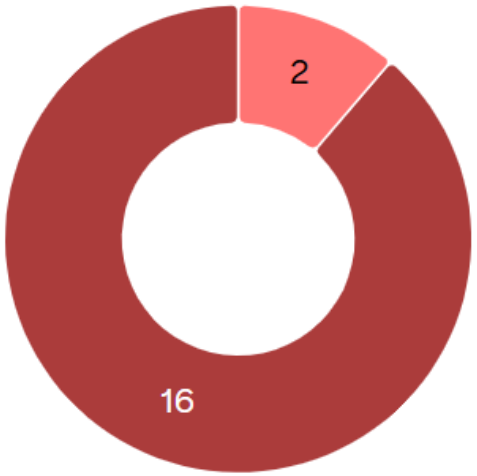

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- $t_{e=0}$ = w
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Propaganda technology becoming better is correct. This would increase incentives to increase propaganda and end up in payoff #2

About last class

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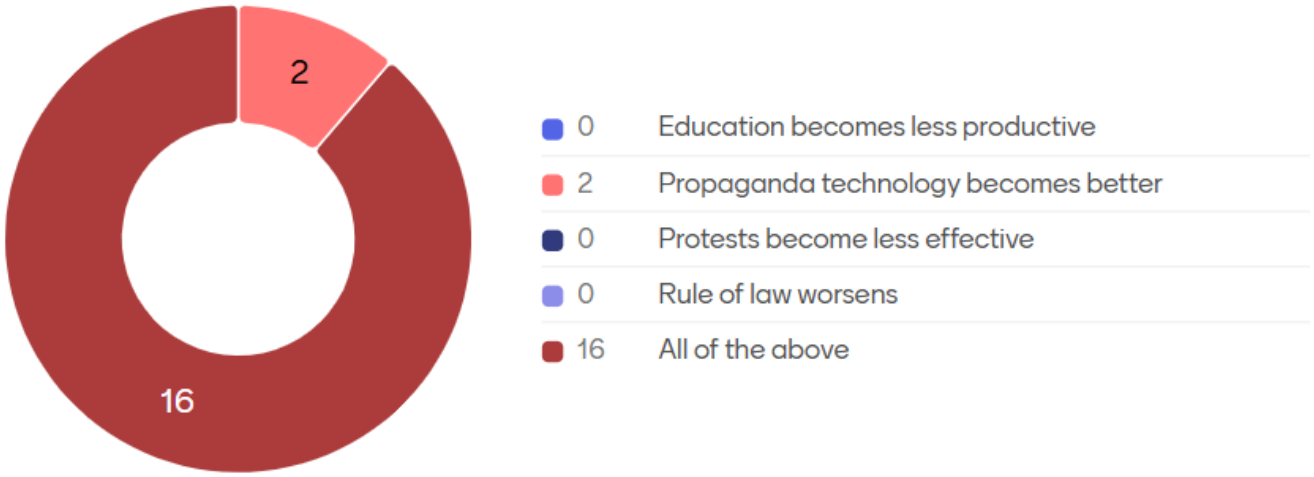

- $t_{e=0}$ = w
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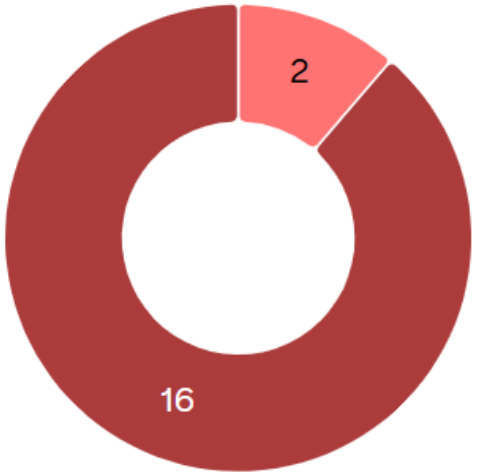
- $t_{c=1}$ = (1- qσ)(w + αh – αc)

Protests becoming less effective would decrease probability of success, q, which would increase payoff #3, so less incentive to invest in propaganda. Similarly, worsening rule of law would decrease σ which would also increase payoff #3, so less incentive to invest in propaganda.

About last class

What could explain this break in trend according to the model?



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- $t_{e=0}$ = w
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As such, “propaganda tech becoming better is the correct answer.

About last class

Taxes are not the only way to extract rents (i.e., corruption)

It is possible to have low taxation and huge rent extraction:

- Privatization corruption (selling public goods at below market value)
- Tax exemptions granted selectively
- Regulatory capture (corporations influence the government to hold monopolies, favourable regulations, special licenses)
- Public procurement corruption (overpayment for public contracts)

Outline for today

- Human Capital and Signalling model
  - Model private returns to education

Readings

- Lovenheim & Turner (2018) Chapter 4 – Human capital model
- Lovenheim & Turner (2018) Chapter 5 – The signalling model: An alternative to the human capital framework

Why are there private returns to education?

In our second class we learned that there are private returns to education – that are both financial and non-monetary

Today we will cover two economic models that attempt to explain why those gains exist:

- Human capital theory: schooling increases human capital which increases productivity
- Signalling model: schooling is a way for high ability individuals to signal their ability

Why are there private returns to education?

- **Human capital theory:** schooling increases human capital which increases productivity
- Signalling model: schooling is a way for high ability individuals to signal their ability

Human capital model

Human capital: set of skills and attributes that individuals have that are productive in the labour market

Human capital can be: skills, health, work-ethic, networks, ideas, knowledge, information, etc.

Human capital model

The idea of the model is that human capital increases productivity and hence wages

- $w_{it}(H_{it}) = \beta_t H_{it}$
- $w_{it+1}(H_{it+1}) = \beta_{t+1} H_{it+1}$
- …

H_{it} is the stock of human capital and β measures the impact of H on wages

Human capital model

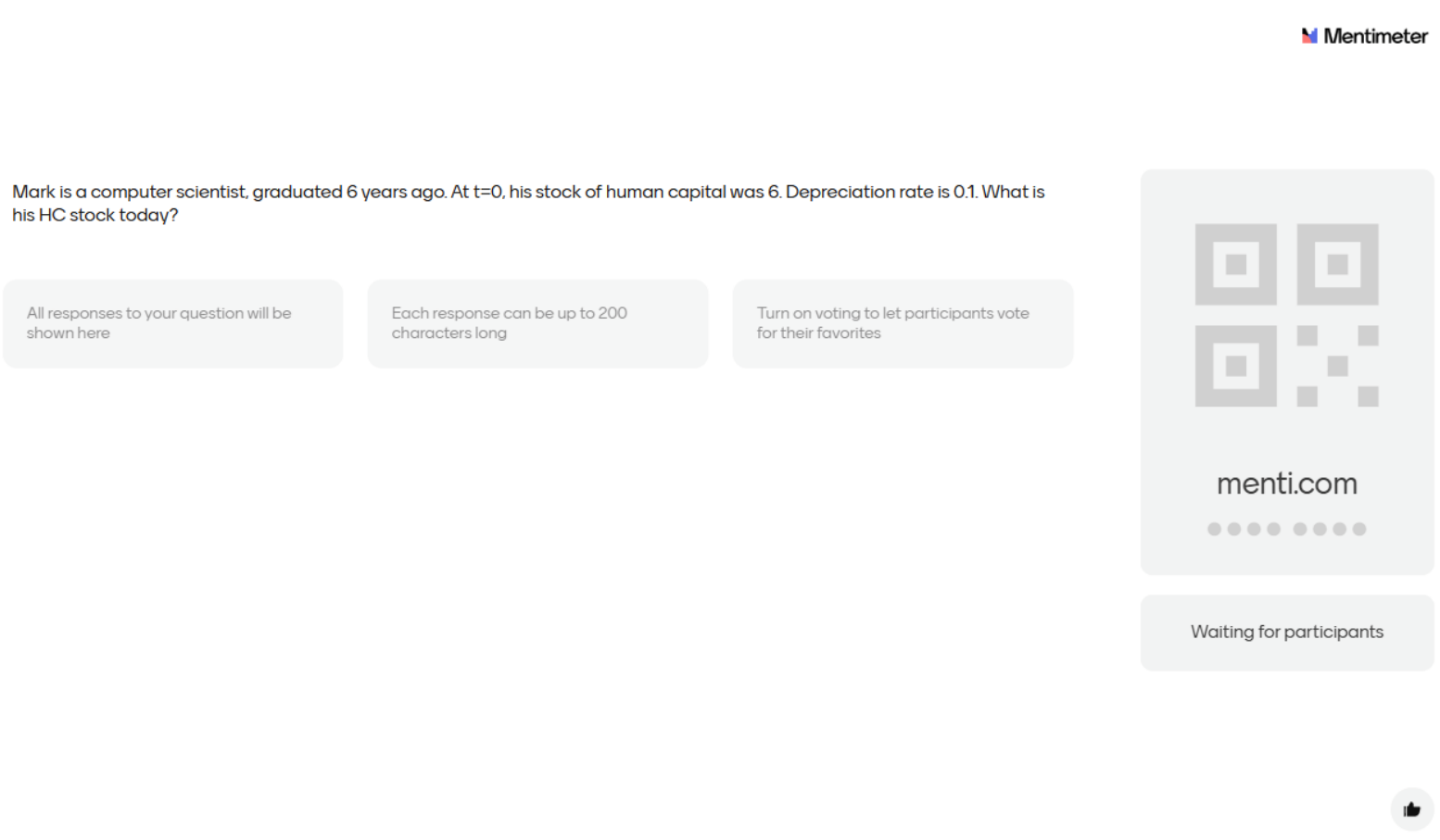
Human capital is a stock; it accumulates from year to year:

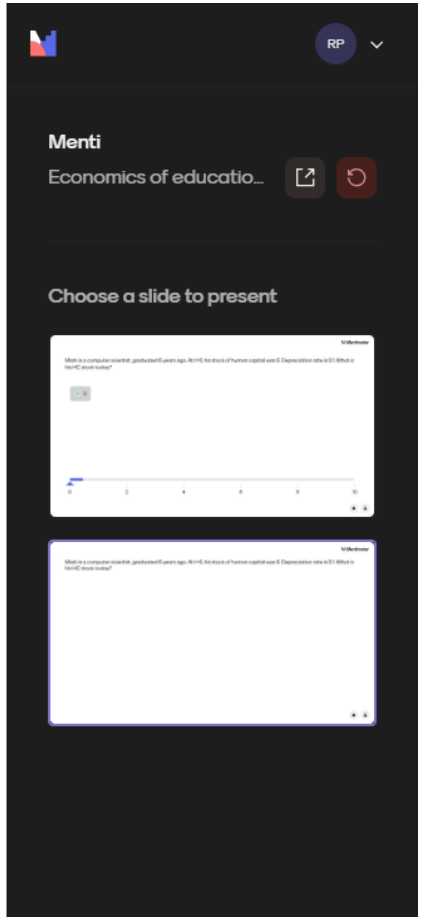
- Education increases the stock
- Depreciation decreases the stock

$$H_{it+1} = H_{it}(1 - \delta) + E_{it}$$

Depreciation Education

Quizz


Mentimeter
Mark is a computer scientist, graduated 6 years ago. At t=0, his stock of human capital was 6. Depreciation rate is 0.1. What is his HC stock today?
All responses to your question will be shown here
Each response can be up to 200 characters long
Turn on voting to let participants vote for their favorites
menti.com
Waiting for participants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Menti
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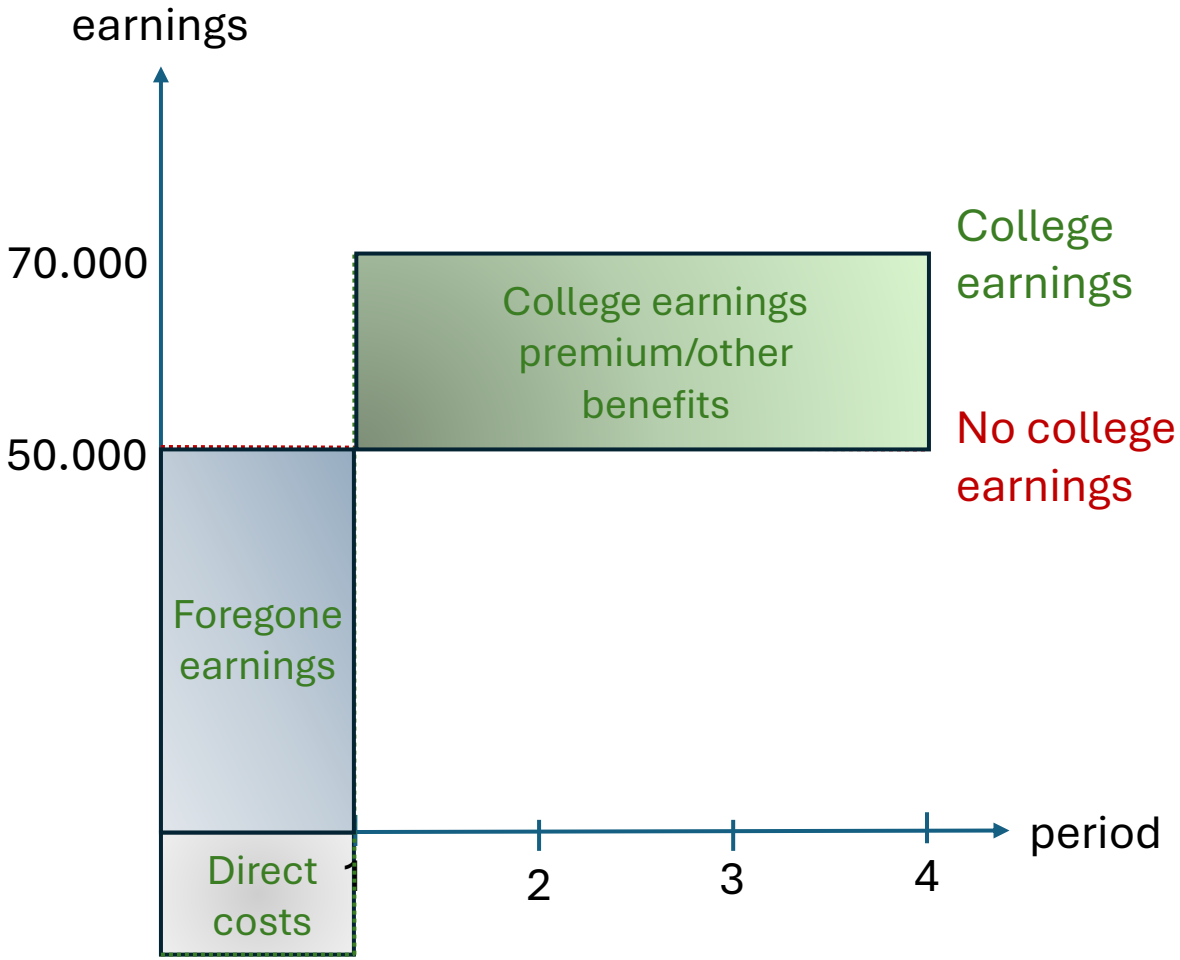
Human capital model

Individuals will invest in human capital if:

- **Present Value (PV)** of schooling is larger than **PV** of no schooling
- <u>or</u> **Net Present Value (NPV)** is larger than 0
- Assumption: no liquidity constraints



Present value

100 euros today

- Worth 100*(1+i) in 1 year
- Worth 100*(1+i)(1+i) in 2 years
- Worth 100*(1+i)(1+i)(1+i) in 3 years
- ....

100 euros today is not the same as 100 euros in 3 years

Costs and benefits should be weighted in same time point, usually the present

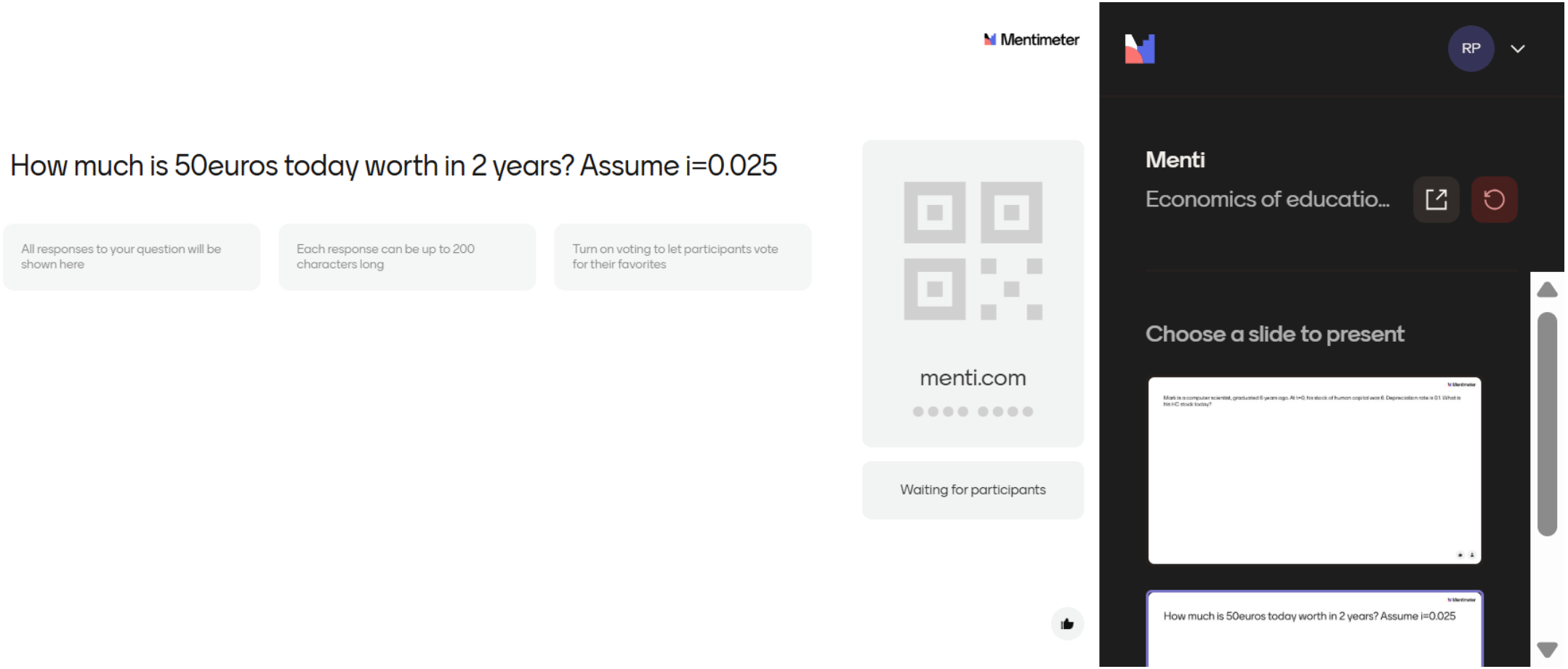
Present value

How much is 100 euros in 3 years worth in present value?

- $100/(1+i) * (1+i) * (1+i)$
- Assuming $i = 0.025$
- 100 euros in 3 years = 92.86 euros today

Present Value (PV) = Future Value ÷ (1 + Discount Rate) ^ Number of Periods

Quizz


Mentimeter
How much is 50euros today worth in 2 years? Assume i=0.025
All responses to your question will be shown here
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RP
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Choose a slide to present
How much is 50euros today worth in 2 years? Assume i=0.025

Human capital model

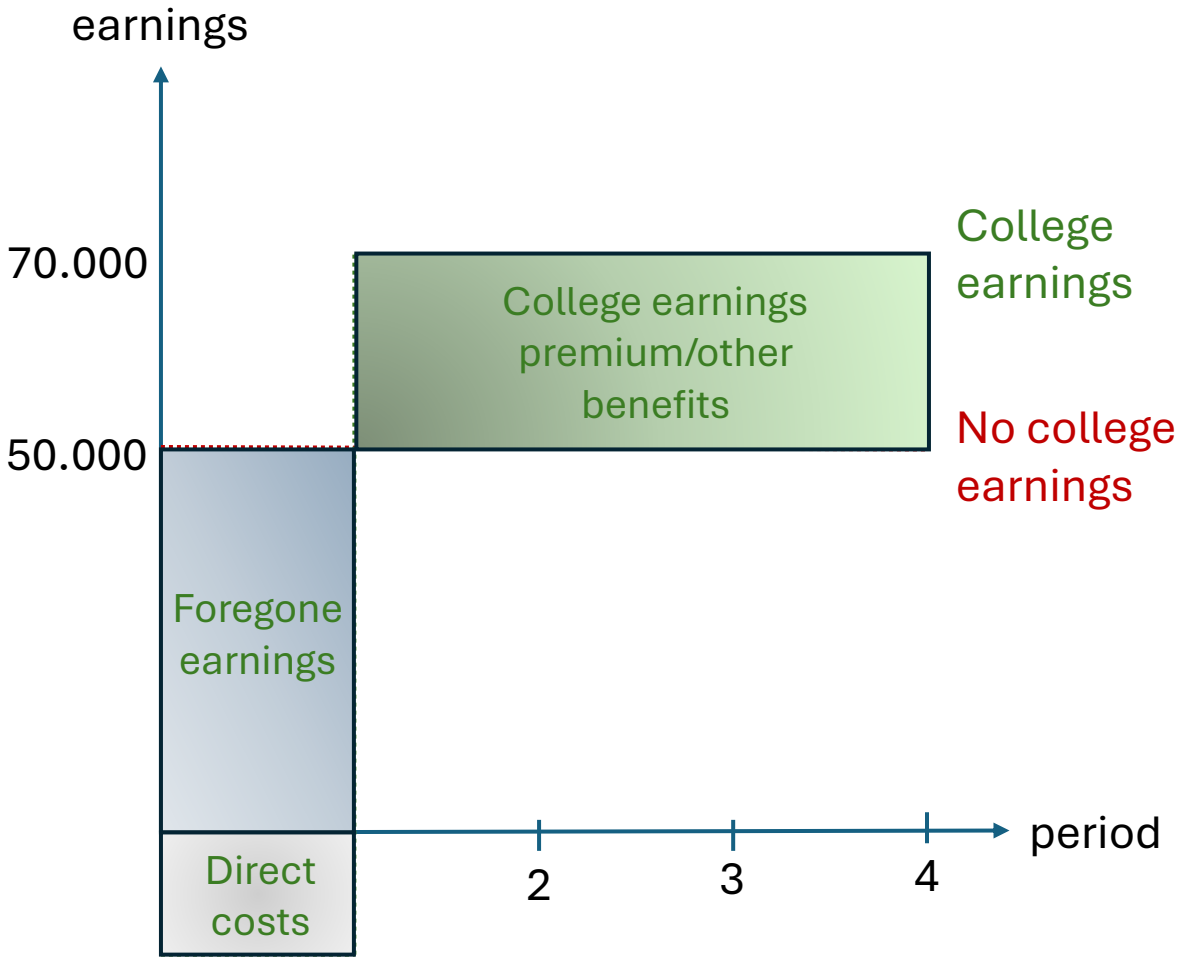
Present value of **no** education:

$$PV(ns) = \frac{50k}{(1+i)} + \frac{50k}{(1+i)^2} + \frac{50k}{(1+i)^3} + \frac{50k}{(1+i)^4}$$

Present value of education:

$$PV(s) = -\frac{20k}{(1+i)} + \frac{70k}{(1+i)^2} + \frac{70k}{(1+i)^3} + \frac{70k}{(1+i)^4}$$

(assumption: payments are done in the end of period)



Human capital model

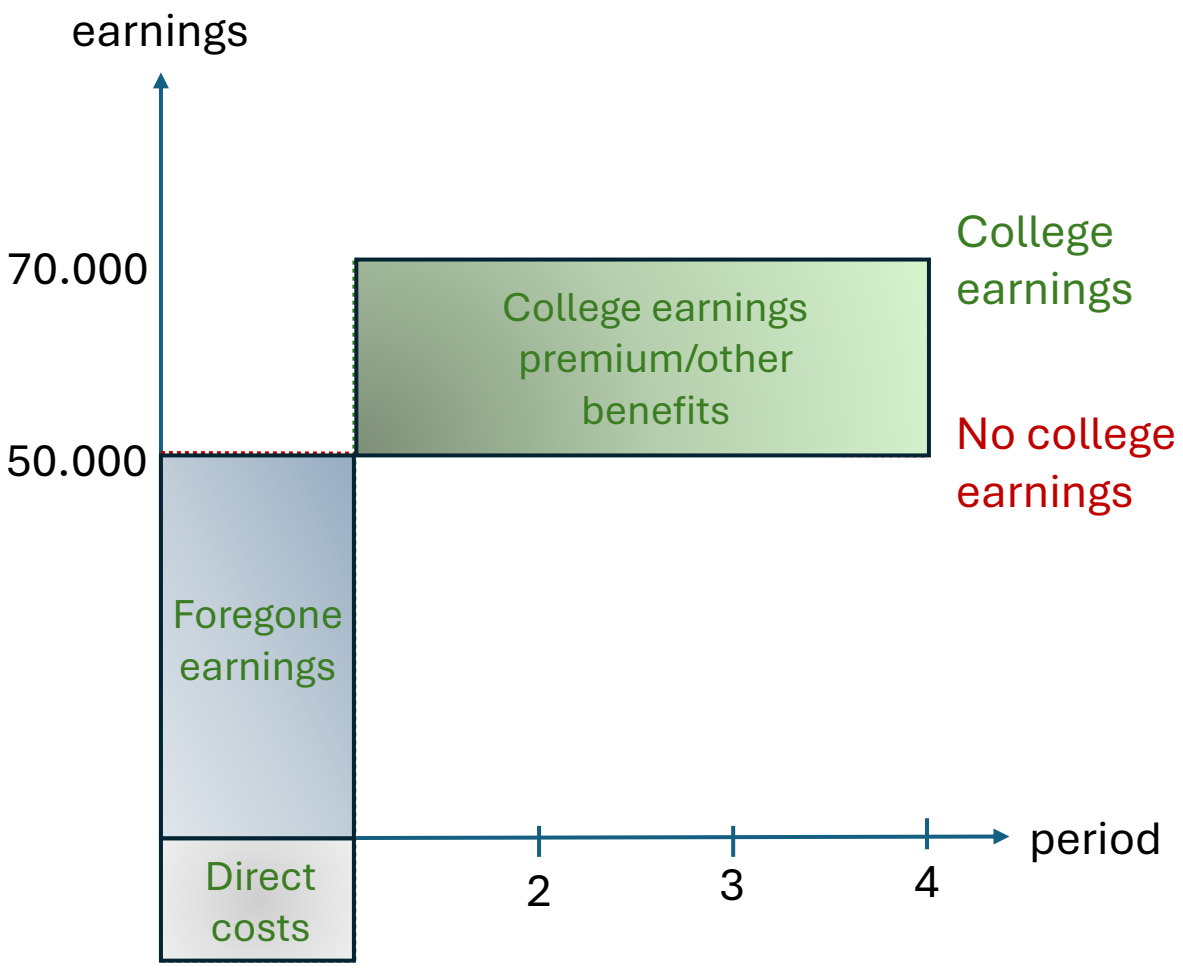
Net present value of education:

$$NPV(s) = PV(s) - PV(ns) =$$

$$= -\frac{20k}{(1+i)} + \frac{70k}{(1+i)^2} + \frac{70k}{(1+i)^3} + \frac{70k}{(1+i)^4}$$

$$-\frac{50k}{(1+i)} - \frac{50k}{(1+i)^2} - \frac{50k}{(1+i)^3} - \frac{50k}{(1+i)^4}$$

$$= -\frac{70k}{(1+i)} + \frac{20k}{(1+i)^2} + \frac{20k}{(1+i)^3} + \frac{20k}{(1+i)^4}$$



Human capital model

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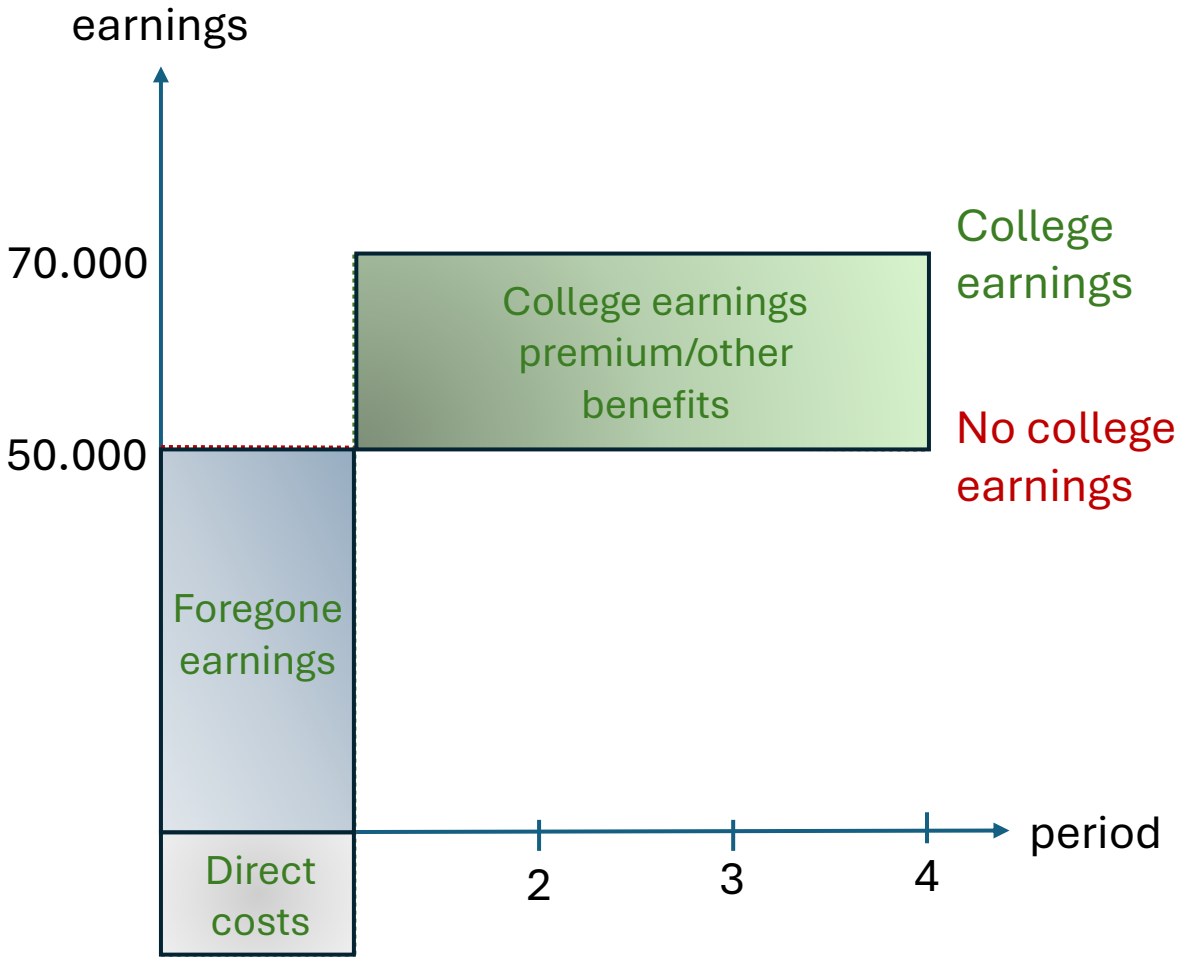
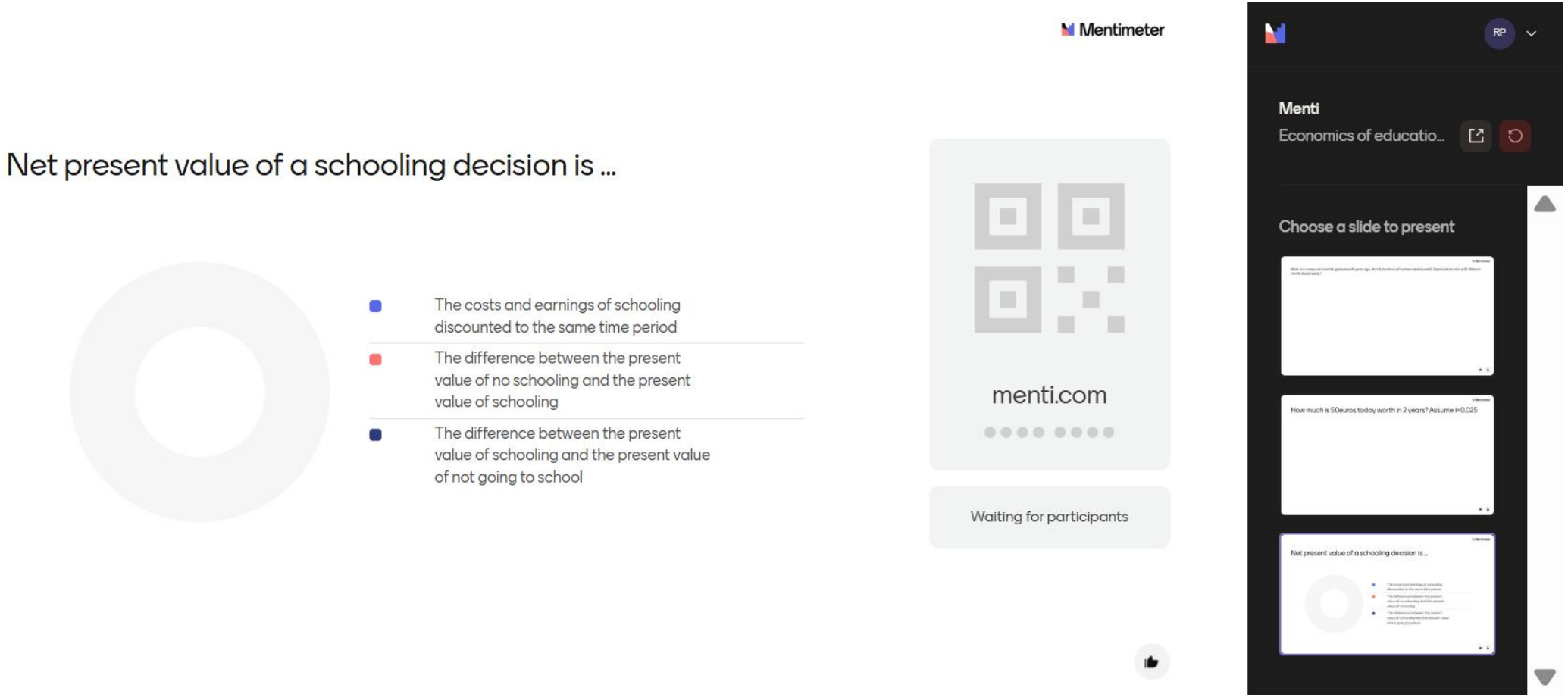
Direct costs
+ opportunity costs

College earnings premium



Quizz


Mentimeter
Net present value of a schooling decision is ...
The costs and earnings of schooling discounted to the same time period
The difference between the present value of no schooling and the present value of schooling
The difference between the present value of schooling and the present value of not going to school
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Choose a slide to present
How much is 50euros today worth in 2 years? Assume i=0.025
Net present value of a schooling decision is ...

Predictions from the model

- Increases in **direct costs** of education (tuition, books, rent increase, etc) **decreases** education
- Increase in **opportunity costs** (wage without education) **decreases** education
- Increases in **education premium increases** education

Continuous schooling choices

- Choice is continuous
- Model assumes diminishing returns to an additional year of education

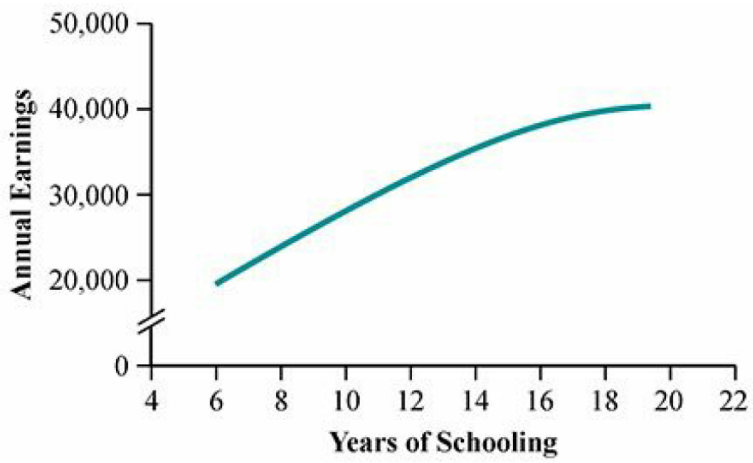

Figure 4.2
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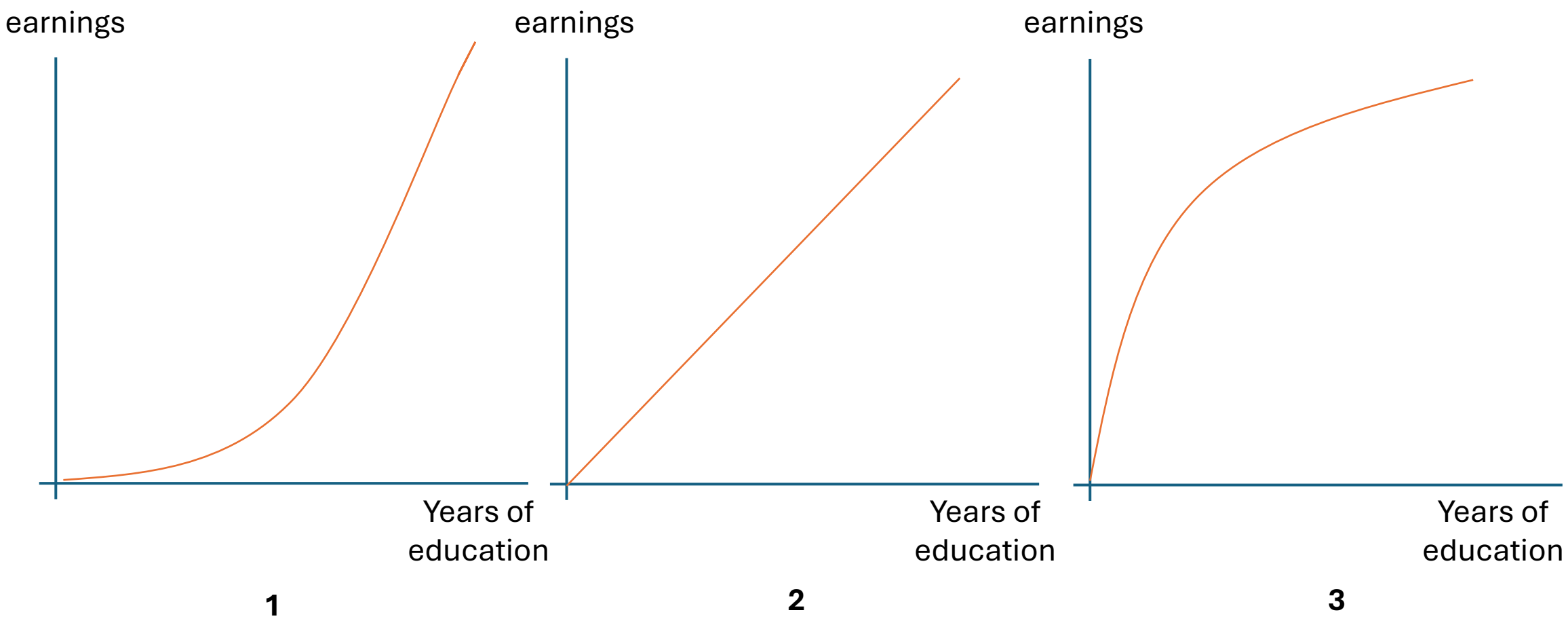
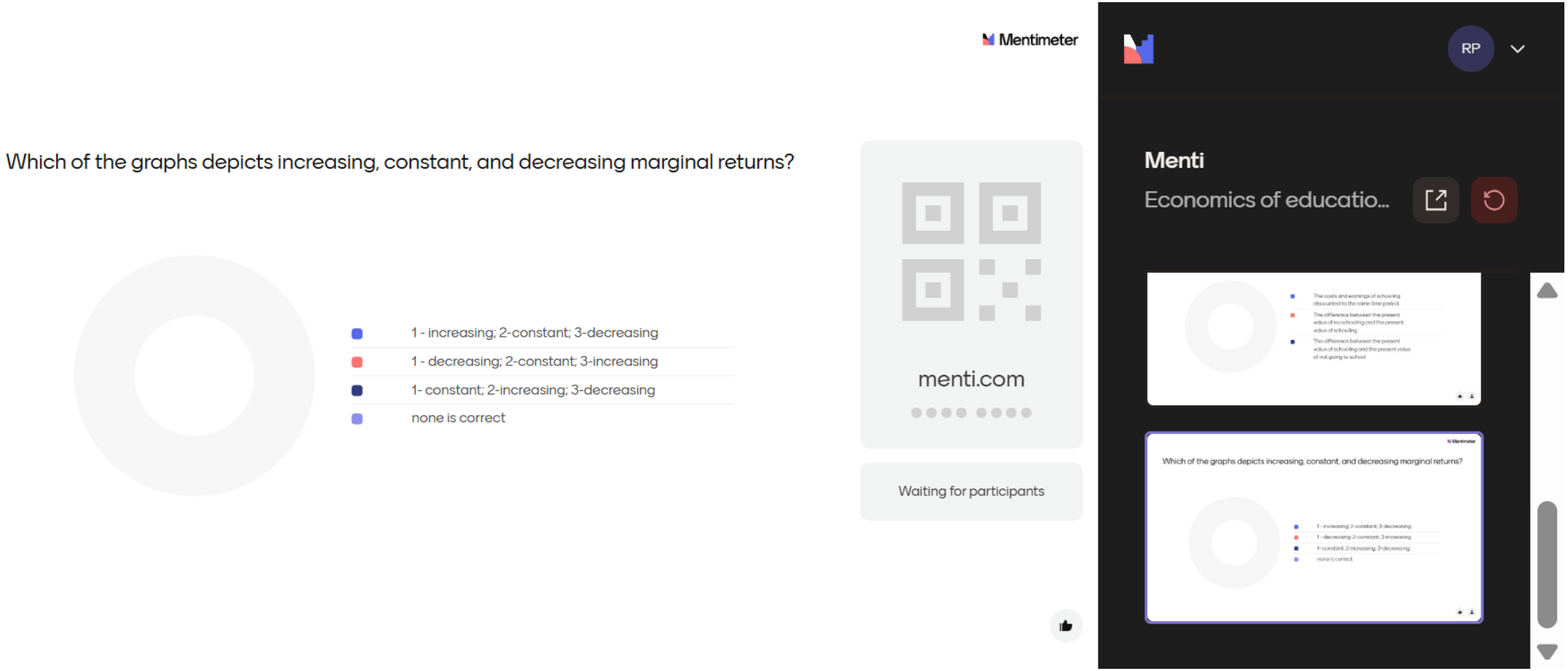
Figure 4.2 The Earnings–Schooling Locus

The earnings–schooling locus shows how expected annual earnings change with years of schooling. It is upward sloping, reflecting the fact that human capital obtained through schooling increases earnings. But it is increasing at a decreasing rate because of the diminishing marginal returns to schooling.

Marginal returns



Quizz


Mentimeter
Which of the graphs depicts increasing, constant, and decreasing marginal returns?
1 - increasing; 2-constant; 3-decreasing
1 - decreasing; 2-constant; 3-increasing
1- constant; 2-increasing; 3-decreasing
none is correct
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Which of the graphs depicts increasing, constant, and decreasing marginal returns?

Marginal returns

- An additional unit of schooling will increasingly/decreasingly/constantly increase your earnings
- Derivatives of a function serve to calculate the function of marginal returns

Marginal returns

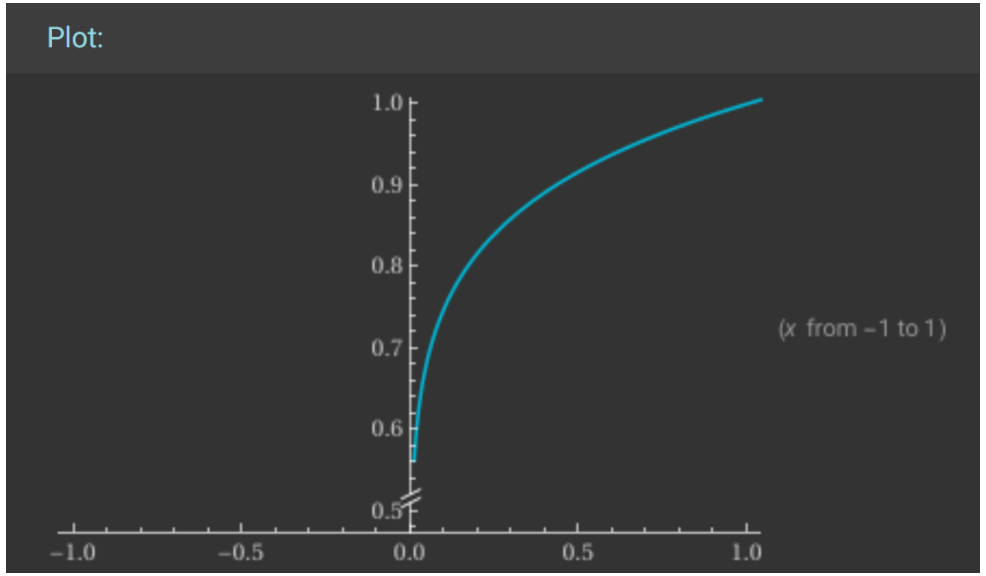
If $e(s) \ = \ s^{1/8}$

Then the derivative, which provides the marginal returns

is $e'(s) \ = \ \frac{1}{8} s^{-7/8}$

Marginal returns

$$e(s) = s^{1/8}$$



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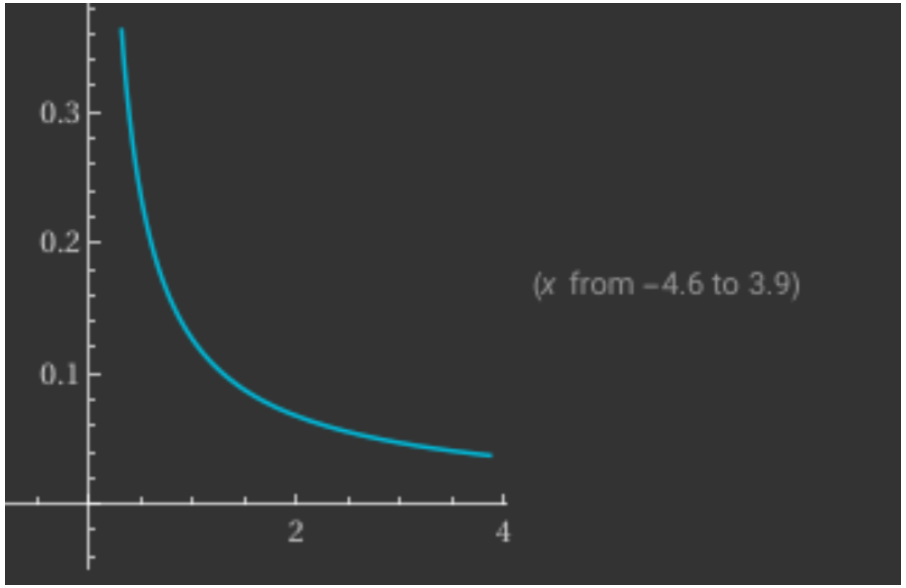

Continuous schooling choices

- Figure depicts diminishing **marginal** returns of education and **marginal** cost of capital
- Assumes a stable interest rate, the cost of borrowing capital to finance education

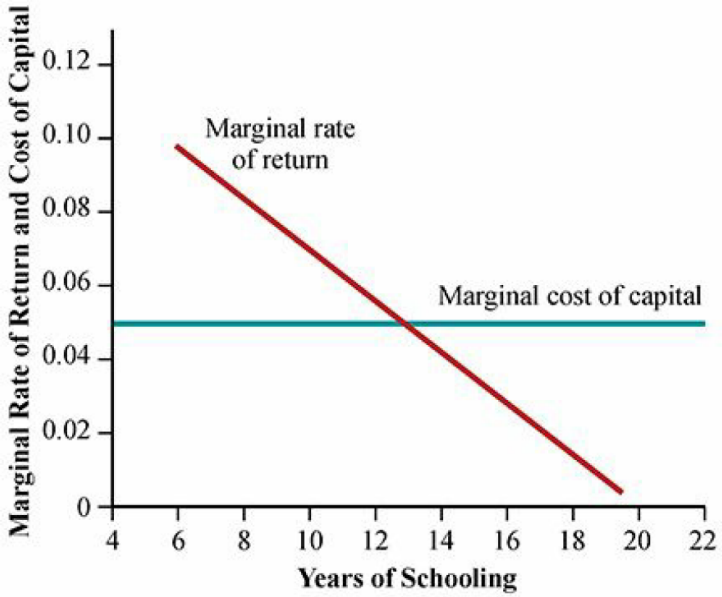

Figure 4.3
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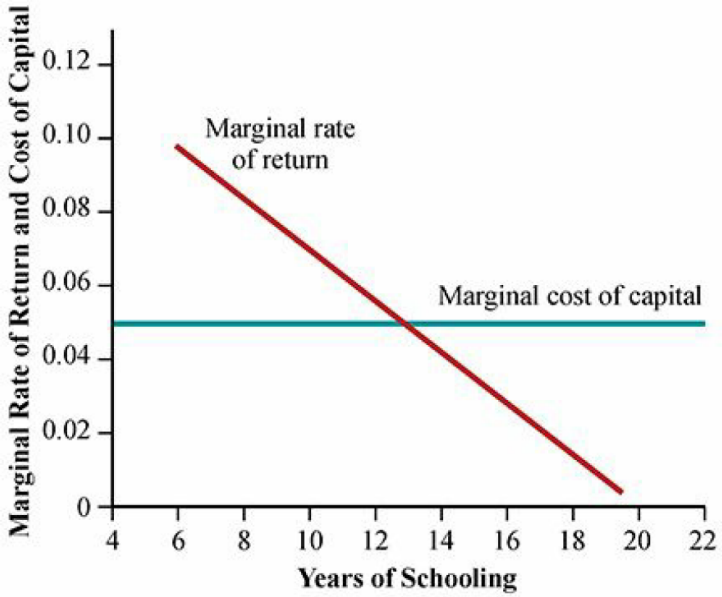
Figure 4.3 Marginal Rate of Return and The Marginal Cost of Capital

The marginal rate of return is downward sloping because of diminishing marginal returns to education. The optimal education investment occurs where the marginal rate of return is equal to the marginal cost of capital.

Continuous schooling choices

In equilibrium, individuals borrow money to stay in school until:

- the marginal cost of borrowing (i) is equal to the marginal benefit of studying (marginal wages)



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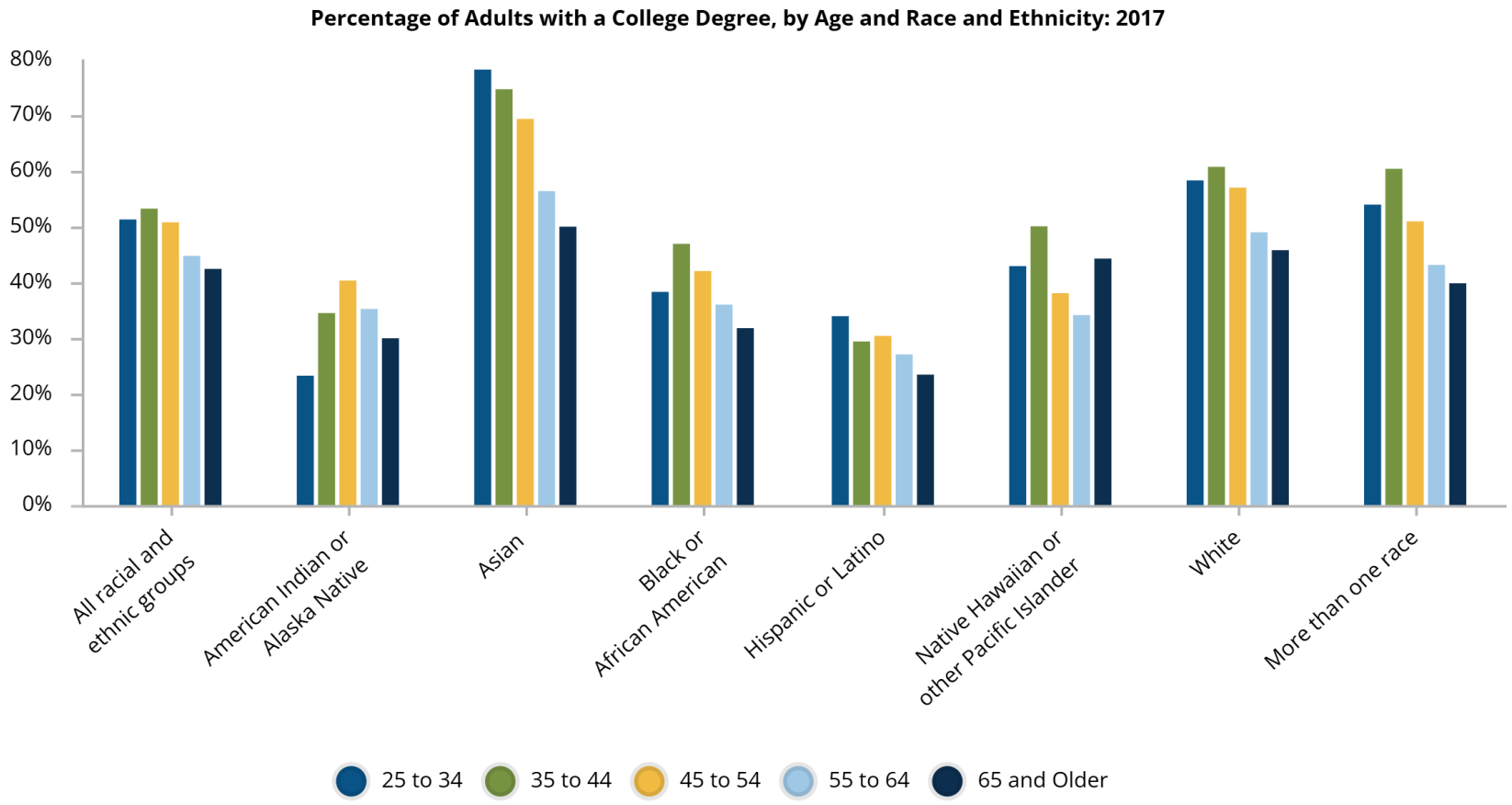
Can the model explain education differences?



Source: U.S. Census Bureau, Current Population Survey, 2022

Can the model explain education differences?

One explanation is that people may differ in their **ability to finance direct costs**

- Not everyone is able to borrow
- If they do borrow, they may be subject to different interest rates
- Conditions of repayment may be different

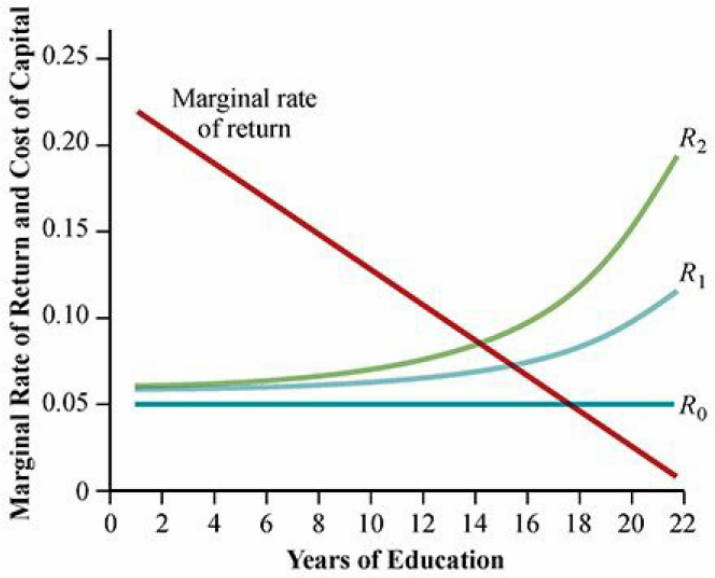


Figure 4.4
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Figure 4.4 Differences in Marginal Cost of Capital Yield Differences in Schooling Investment

There are three cost of capital curves: R_0, R_1, and R_2. R_1 and R_2 are upward sloping because the cost of funds increases as the level of education paid for by borrowing goes up. Students who face more difficulty borrowing have more steeply sloped marginal cost of capital curves and as a result obtain fewer years of education.

Can the model explain education differences?

A second explanation is that people may differ in the **benefits they receive** from education

- People differ in ability and personality
- Two people receiving the same amount of education might gain different levels of human capital

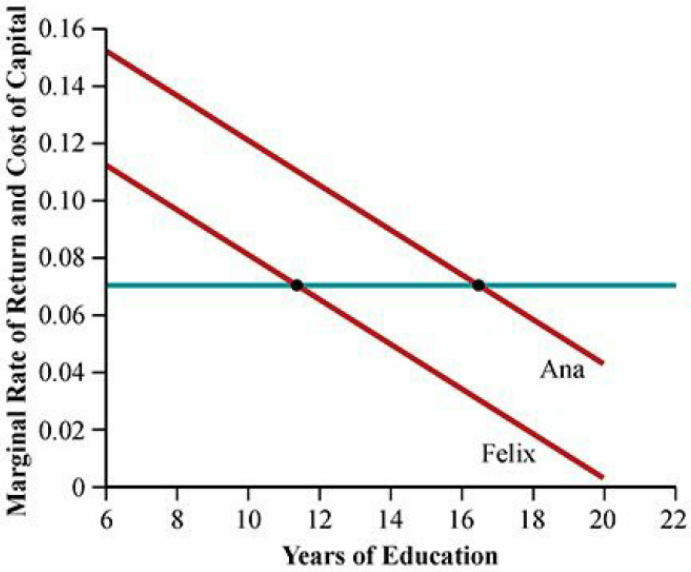

Figure 4.5
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Figure 4.5 Marginal Rates of Return for Workers of Different Ability Levels

Ana is more able than Felix, so her marginal rate of return schedule is above Felix's at every level of schooling. Both Ana and Felix invest in the efficient level of schooling—the point where their marginal rate of return intersects with the marginal cost of capital. Ana obtains more education than Felix because she is of higher ability.

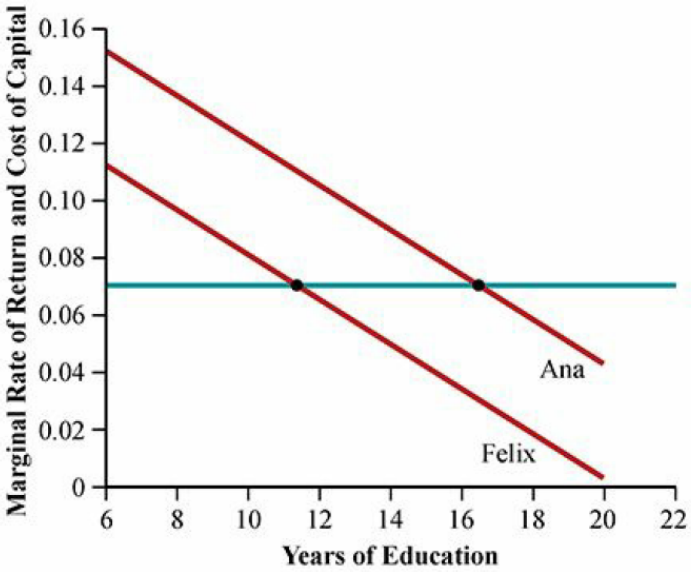
Can the model explain education differences?

Differences in education are explained through:

- Different costs of borrowing
- Different benefits of schooling

Can the model explain education differences?

Could you calculate the rate of return to schooling by subtracting wages of Ana and Felix?



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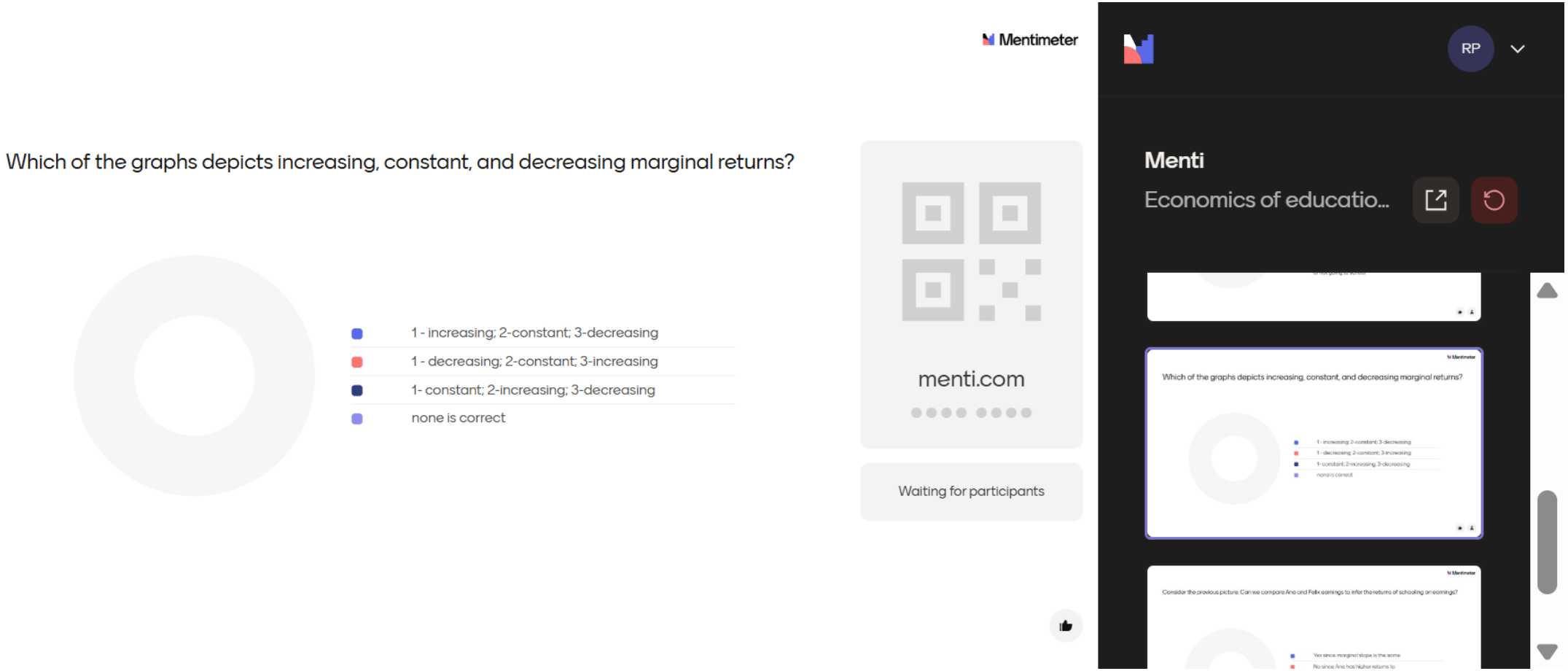
Quizz


Mentimeter
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1 - decreasing; 2-constant; 3-increasing
1- constant; 2-increasing; 3-decreasing
none is correct
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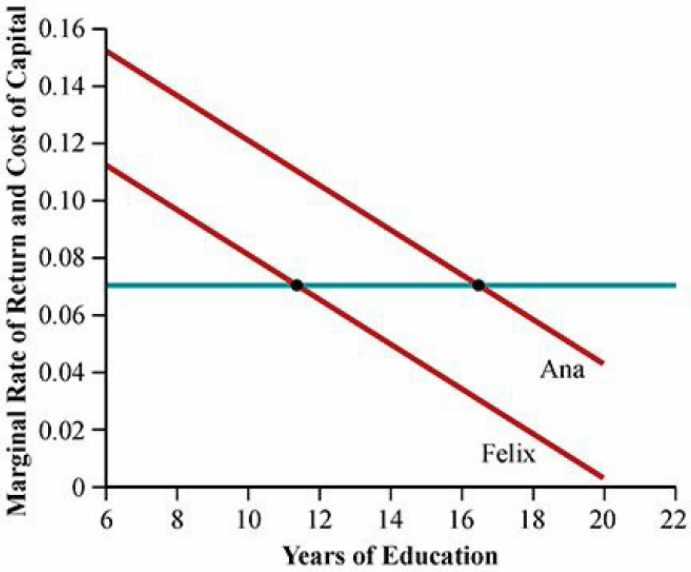
Can the model explain education differences?

Note that the rate of return can also incorporate other nonmonetary benefits or costs! Such as:

- Disliking studying
- How easy it is to have good grades
- Being close to best friend in the same school
- …



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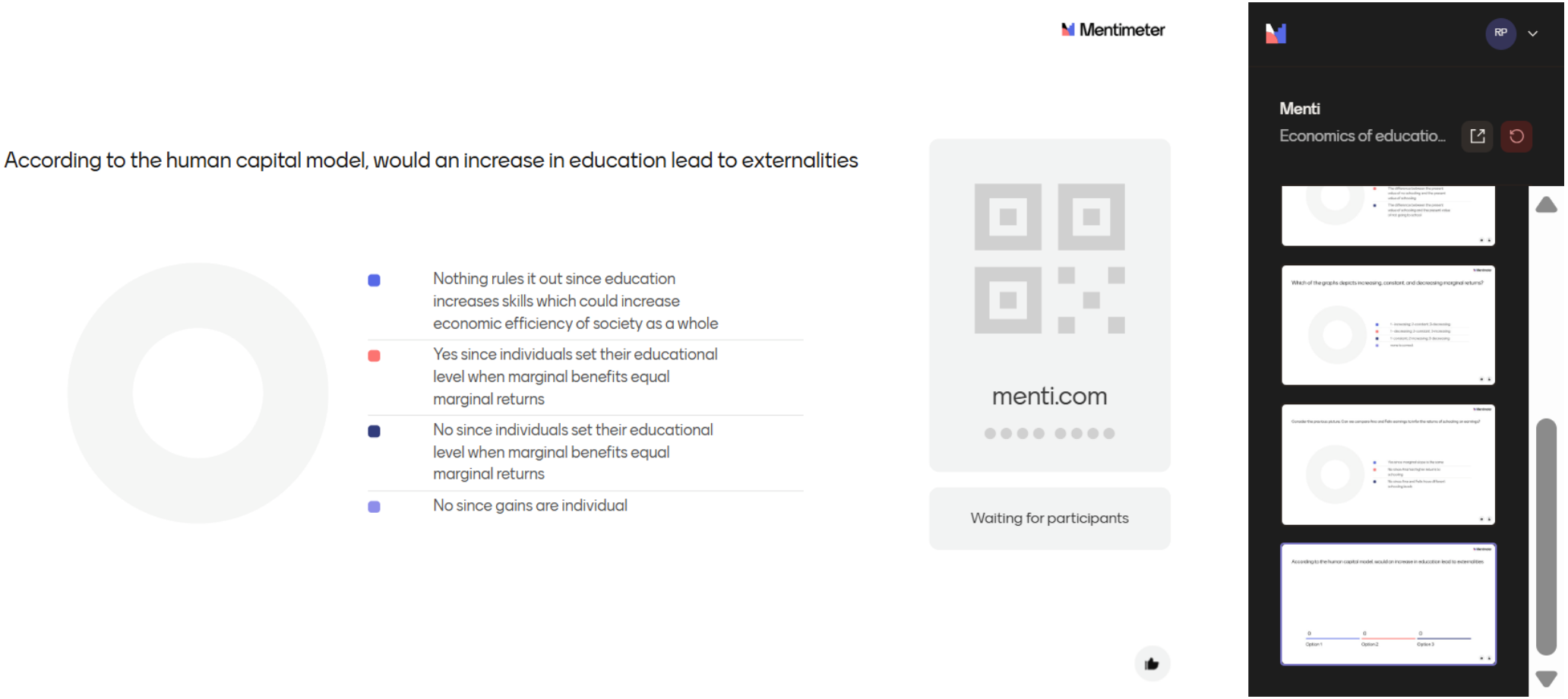
Human capital model take-aways

According to this model:

- Increased costs, reduced education premiums, less years in the work force, or increasing interest rates decreases years education
- People differ in their characteristics such that they face different marginal benefits and marginal costs from investing in education
- People may also differ in access to credit

Quizz


Mentimeter
According to the human capital model, would an increase in education lead to externalities
Nothing rules it out since education increases skills which could increase economic efficiency of society as a whole
Yes since individuals set their educational level when marginal benefits equal marginal returns
No since individuals set their educational level when marginal benefits equal marginal returns
No since gains are individual
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Economics of educatio...

Why are there private returns to education?

- Human capital theory: schooling increases human capital which increases productivity
- **Signalling model:** schooling is a way for high ability individuals to signal their ability

Signalling model

- The hypothesis of this model is that higher levels of education are associated with higher productivity because **education signals or screens high productivity** and not because education raises productivity

Spence, M. (1978). Job market signaling. In *Uncertainty in economics* (pp. 281-306). Academic Press.

Stiglitz, J. E. (1975). The theory of" screening," education, and the distribution of income. *The American economic review*, *65*(3), 283-300.

Signalling model

- In economies with imperfect information, individuals with productive “unobservable” characteristics will choose education as a costly signal device
- No signals means an average wage for everyone
- Signals means “productive” individuals will get a higher wage at the expense of people without signals

Spence, M. (1978). Job market signaling. In *Uncertainty in economics* (pp. 281-306). Academic Press.

Stiglitz, J. E. (1975). The theory of" screening," education, and the distribution of income. *The American economic review*, *65*(3), 283-300.

Signalling model

- In general, an activity may be used as a signal of “productivity” if its less costly for someone “productive” than for someone “unproductive”
- So, schooling must be less costly for “productive” individuals

Spence, M. (1978). Job market signaling. In *Uncertainty in economics* (pp. 281-306). Academic Press.

Stiglitz, J. E. (1975). The theory of" screening," education, and the distribution of income. *The American economic review*, *65*(3), 283-300.

Signalling model

Lower costs
These costs reflect ease

Types	*Productivity*	*Costs of schooling*	*Proportion in population*
I	A_I	$\gamma_I=\gamma(S, A_1)$	q
II	A_{II}	$\gamma_{II}=\gamma(S, A_2)$	$1-q$

- A1 is more productive, and acquires education at a lower cost
- If firms are not able to distinguish types, then:

$$w = qA1 + (1-q)A2 = \bar{A}$$

Signalling model

Worker invests in education if benefits of schooling exceed costs

A1: invests in education if $A1 - A2 > \gamma 1$

A2: invests in education if $A1 - A2 > \gamma 2$

Education premium

Signalling model

A1: invests in education if $A1 - A2 > \gamma 1$

A2: invests in education if $A1 - A2 > \gamma 2$

For there to be a signalling equilibrium,

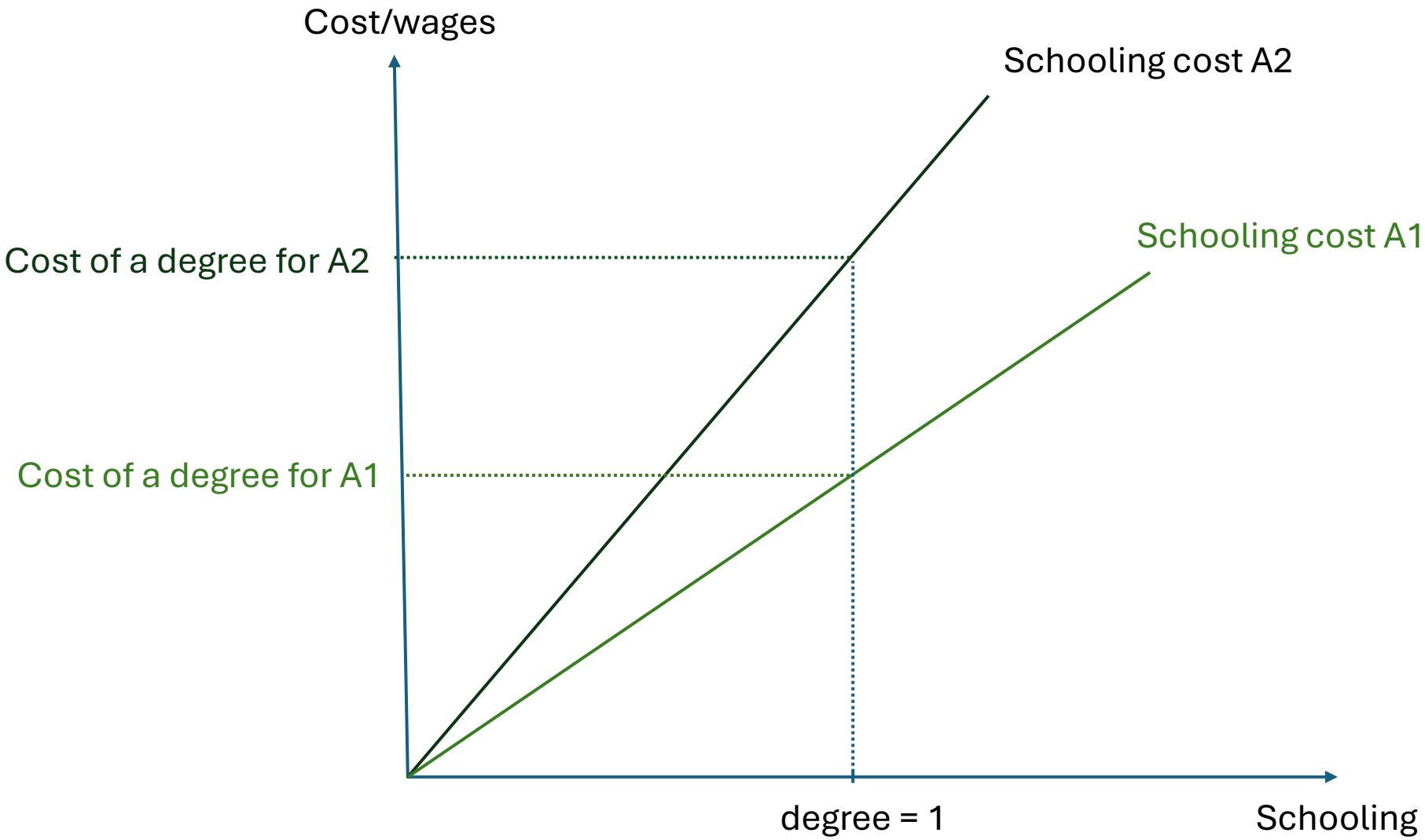
A1 invests $A1 - A2 > \gamma 1$

A2 does not $A1 - A2 < \gamma 2$

So $\gamma 1 < A1 - A2 < \gamma 2$

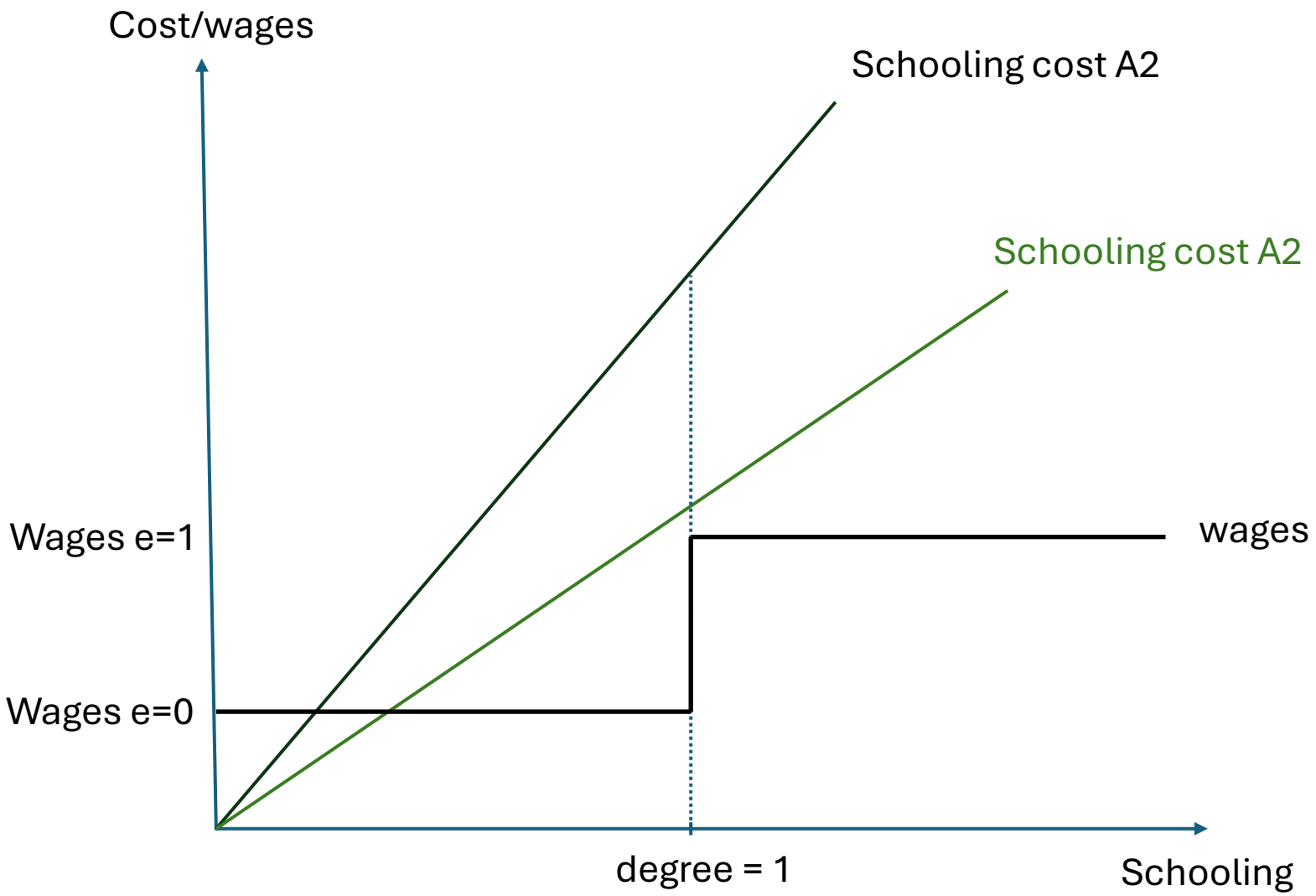
The difference between High and Low productivity wage (A1- A2) must be higher than cost of A2 of obtaining a degree, and lower than the cost of A1 of obtaining a degree

Signalling model



Signalling model

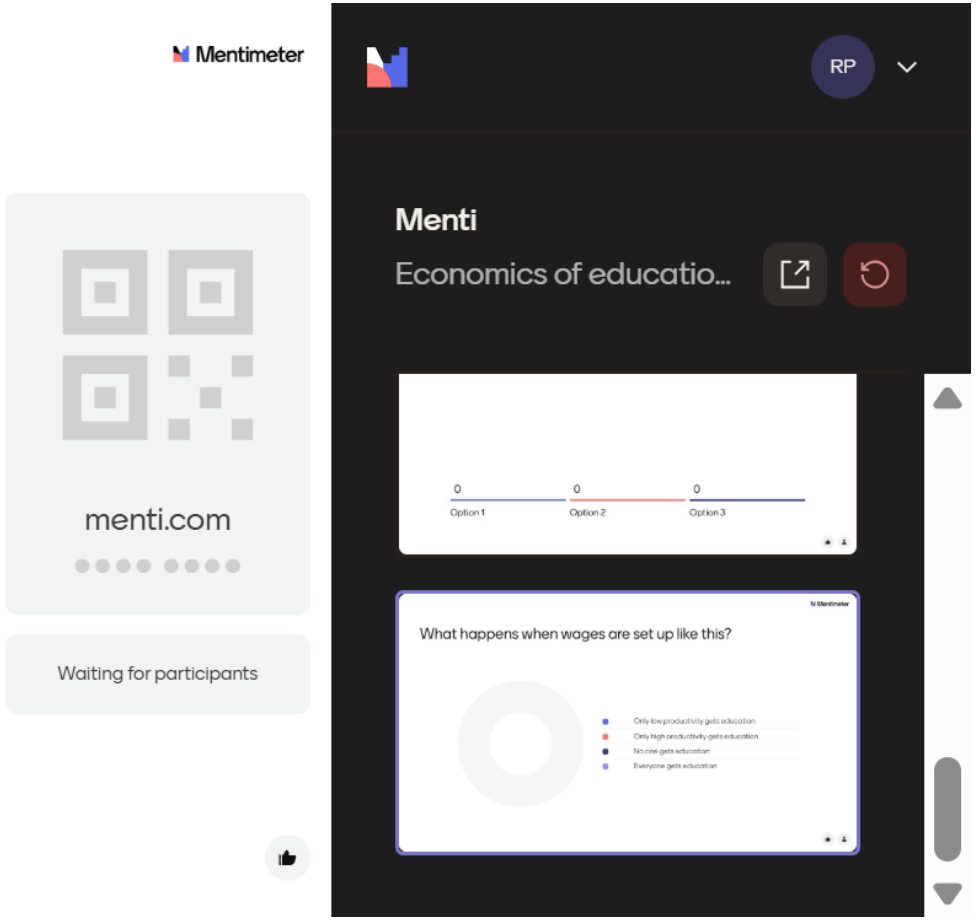
What happens when wages are set up like this?



Quizz

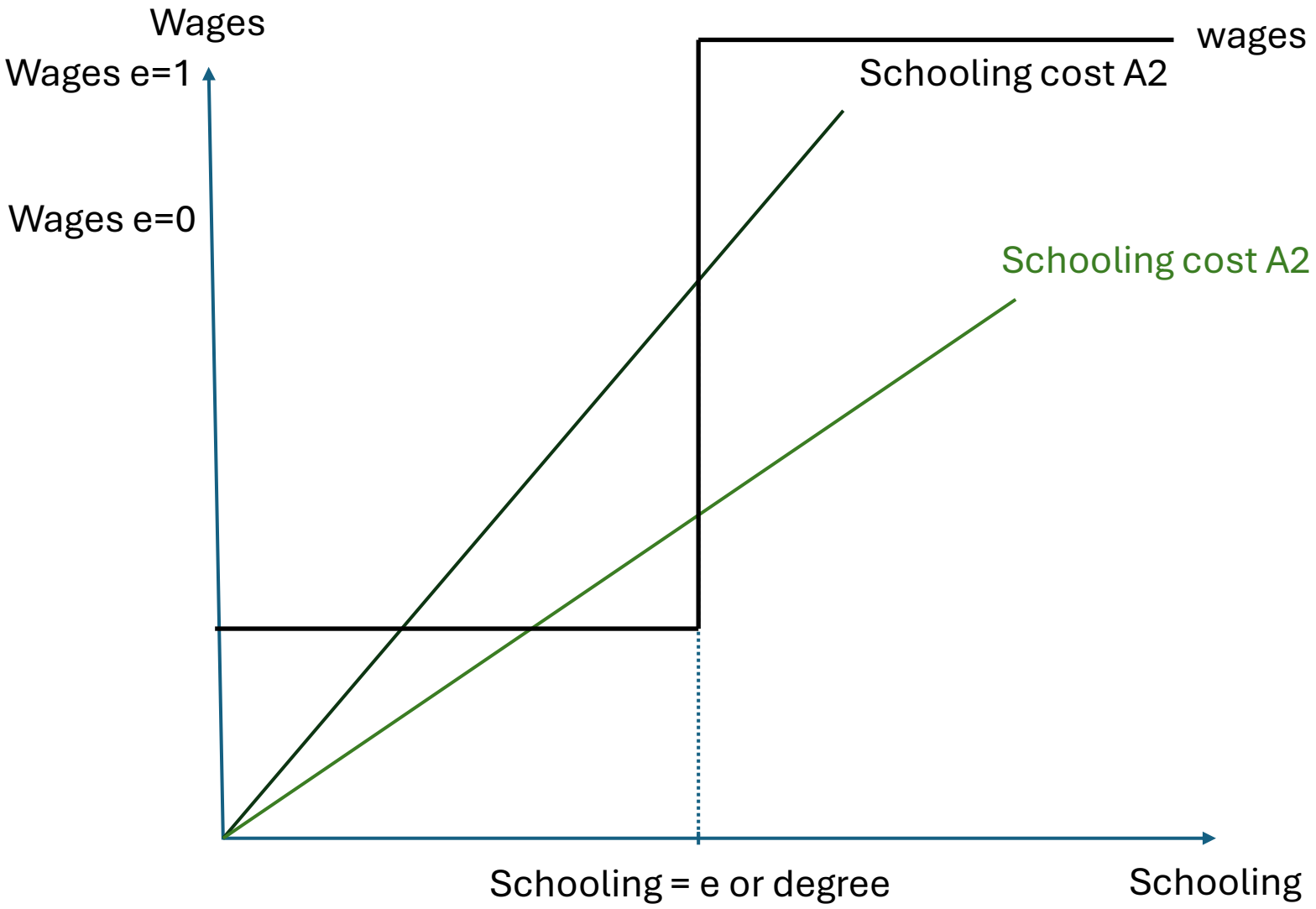
What happens when wages are set up like this?

- Only low productivity gets education
- Only high productivity gets education
- No one gets education
- Everyone gets education

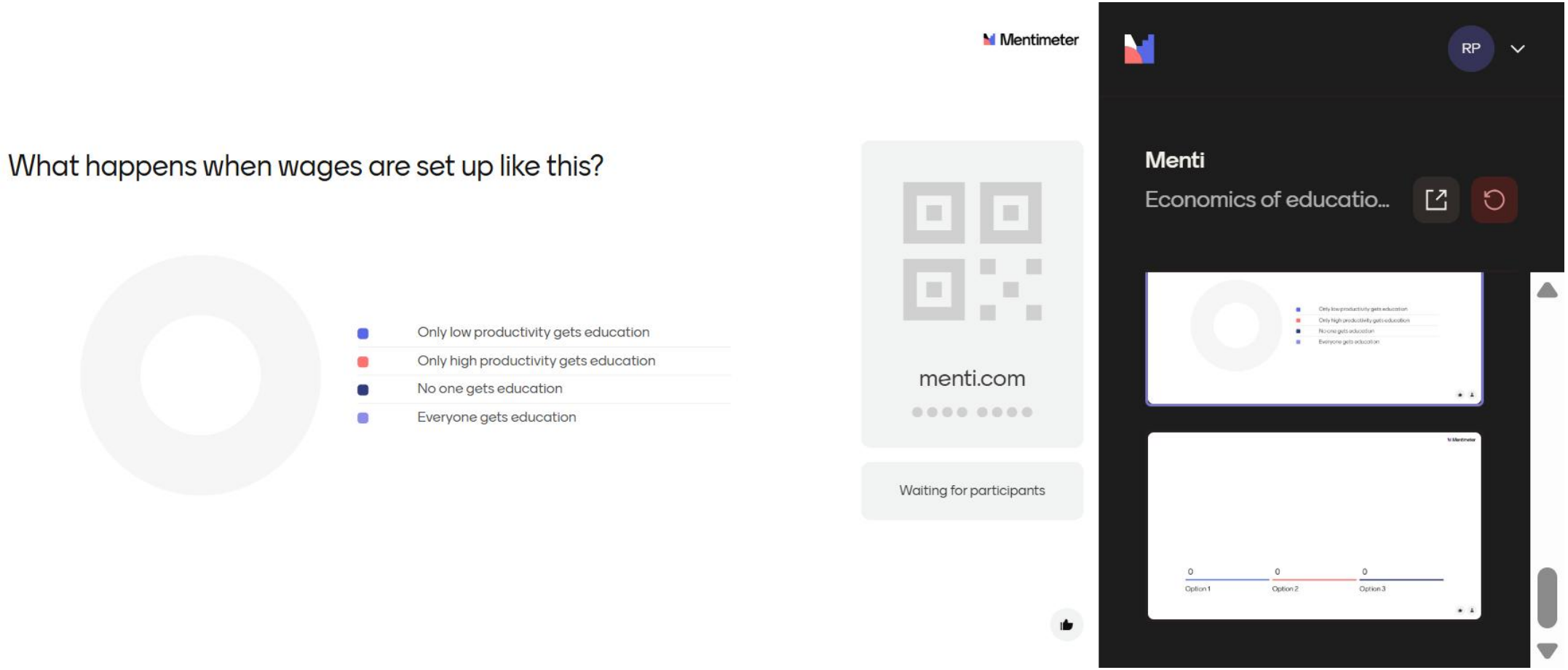


Signalling model

What happens when wages are set up like this?



Quizz


What happens when wages are set up like this?
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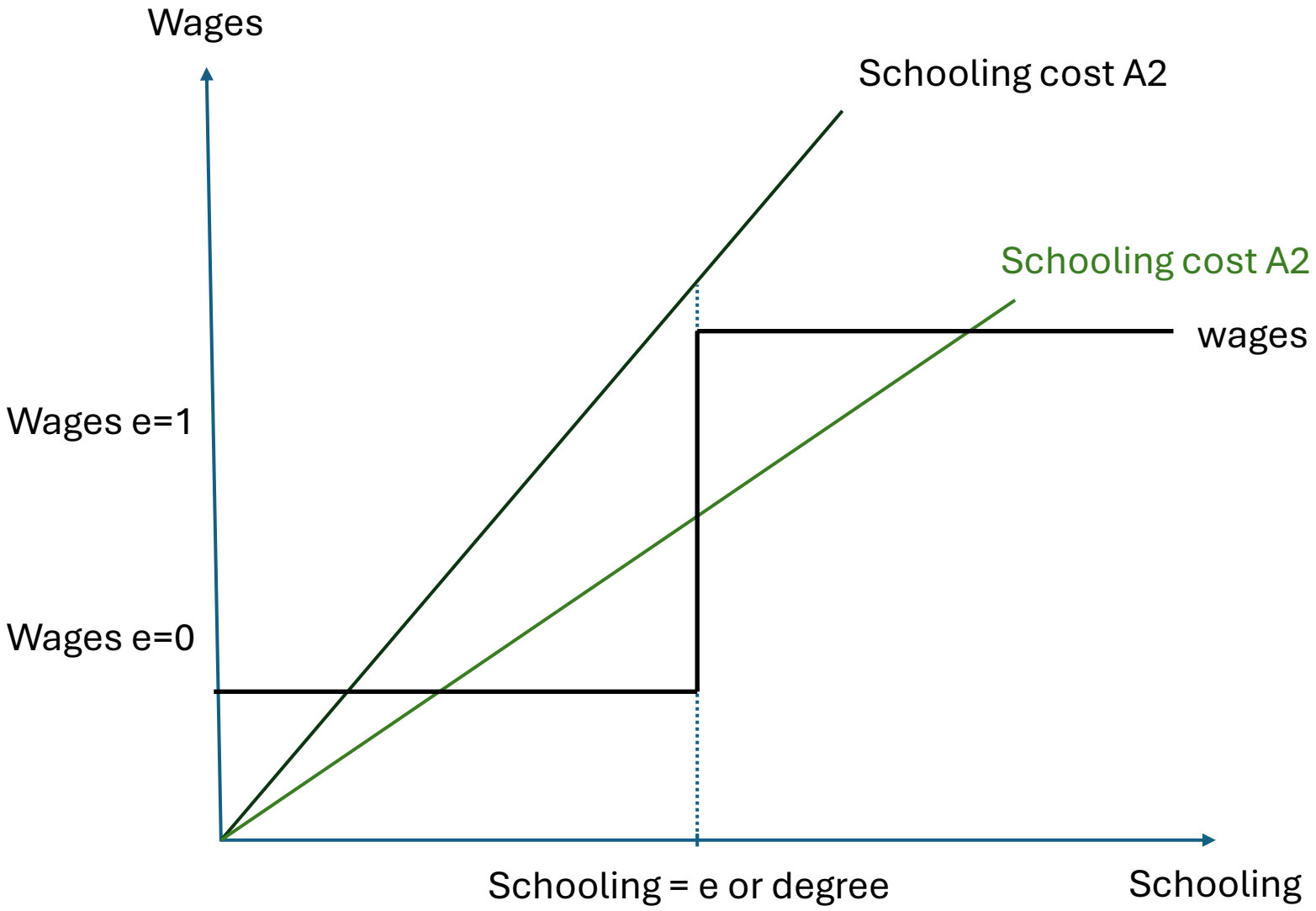
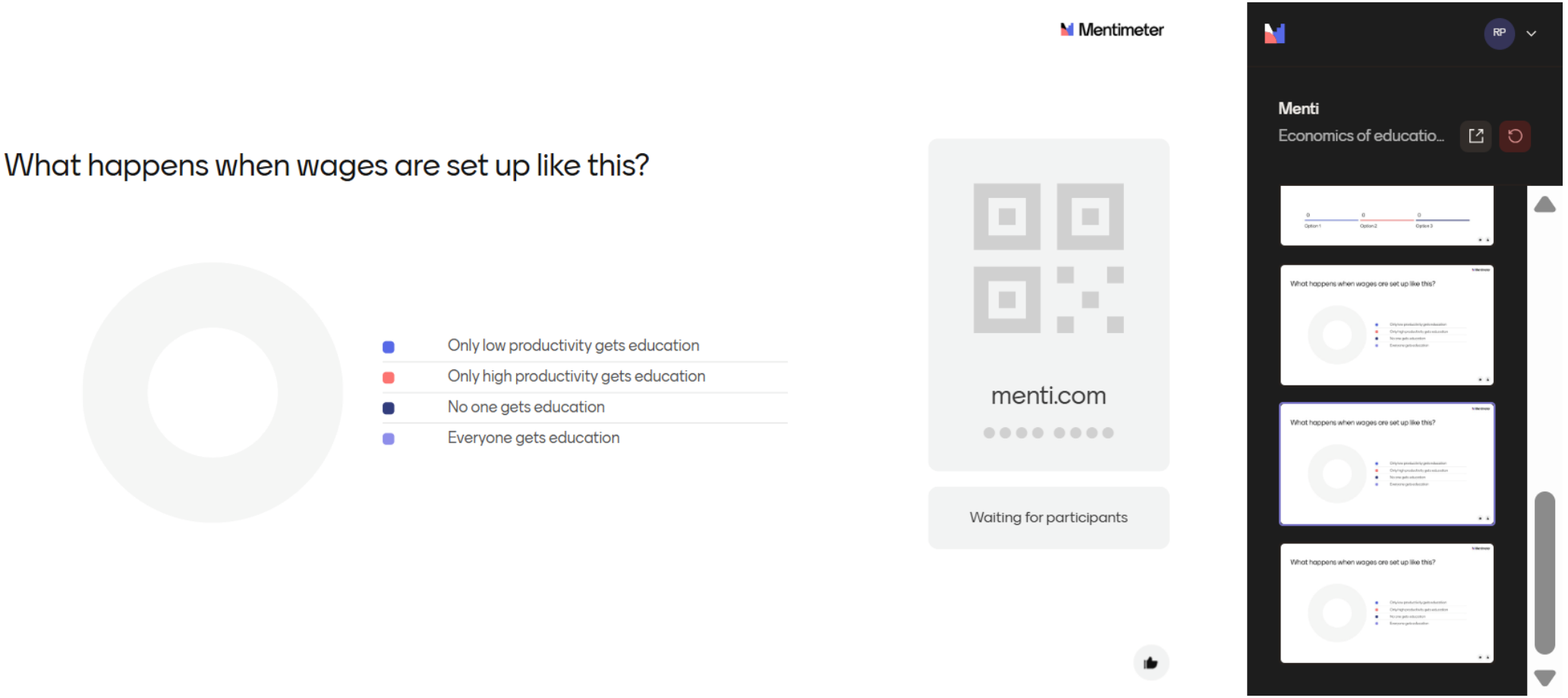
Option 1
Option 2
Option 3

Signalling model

What happens when wages are set up like this?



Quizz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Signalling model

- If degrees are too expensive or difference between wages is too small
  - There could be an equilibrium where no one wants education

- If degrees are very cheap or differences between wages are very large
  - There could be an equilibrium where everyone gets education

Signalling model

Employer pays higher wage if employee has a degree (=e)

If it only pays for high productivity worker to invest in one

Separating equilibrium

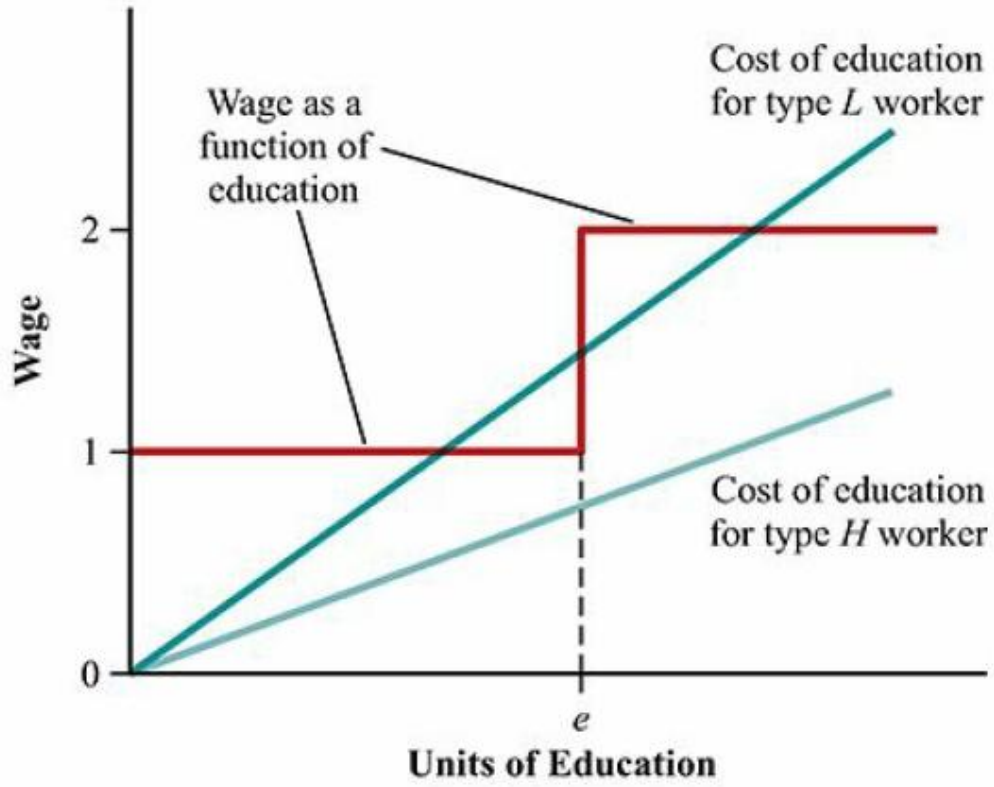

Figure 5.2
Lovenheim/Turner, *Economics of Education*, 1e,
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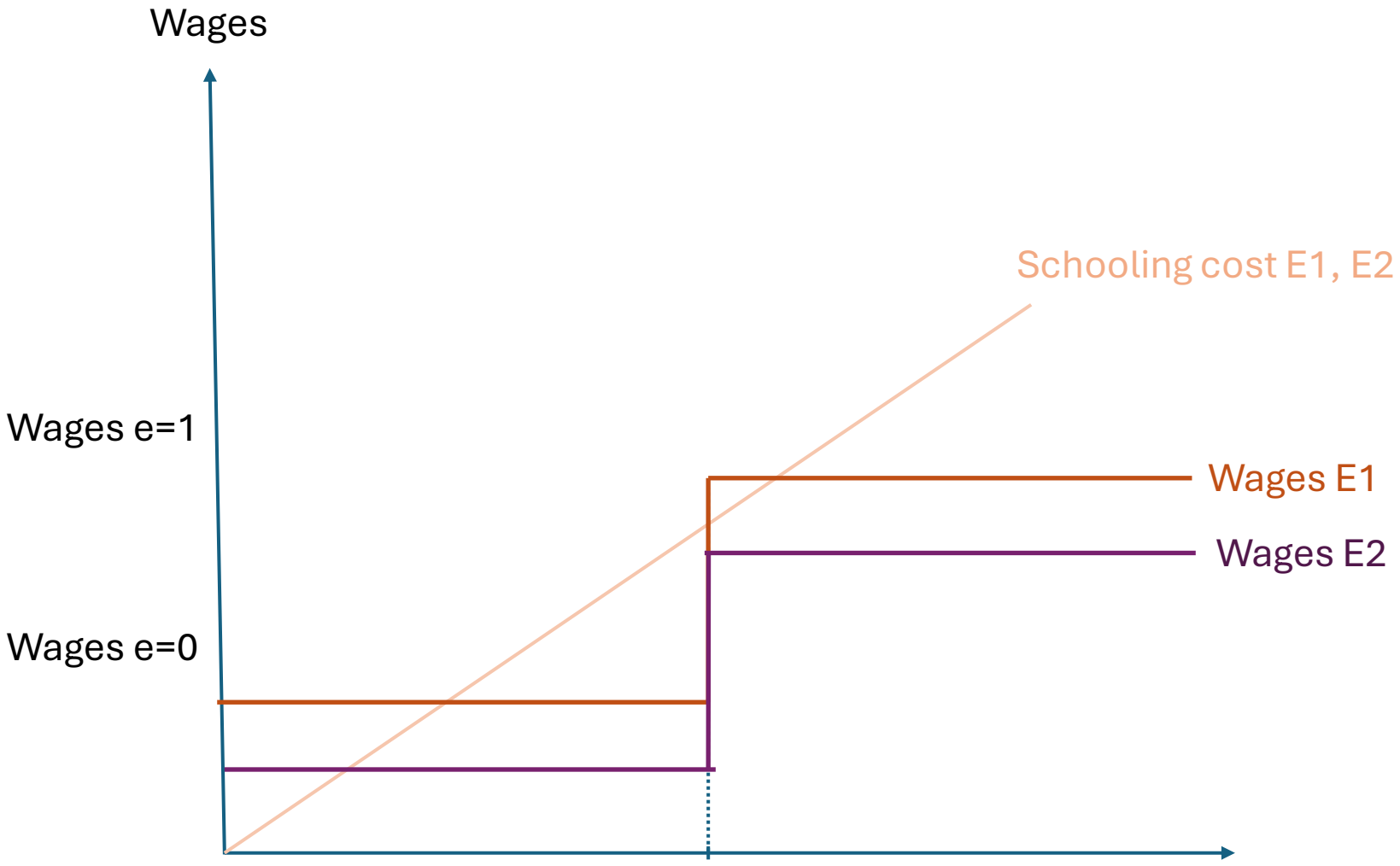
Signalling model – other educational differences

- Employer is biased against certain types (e.g., ethnicity, age, gender,…)
- Employer assumes that certain types have lower productivity
- This changes incentives for individuals that belong to certain types to pursue education
- Their returns are not only dependent on their signal, but on their type

Signalling model

In this situation two people from different ethnicities (E1,E2) have the same schooling costs (= same productivity)

However because the employer discriminates against E2, it is not worth for this person to invest in education since costs are larger than benefits



Signalling model take aways

This model predicts that:

- Investing on education acts as a signal of ability
- There are no positive externalities
- Education is simply a mechanism to have higher wages at the expense of others
- **Education is inefficient**; no productivity gains, it simply taxes highly productive individuals to access higher wages

Take aways (1)

- Human capital model is based on the idea that education generates knowledge and skills
- Signalling model is based on the idea that education is only a signalling device
- Models diverge in the purpose of education for the economy
- Under the HC model, education increases productivity
- Under the signalling model, education only serves as a costly signal device
- Different policy implications in terms of **financing** education

Take aways (2)

- Human capital assumes that differences in education are explained by different **cost of capital** or in **private benefits**
- Signalling model assumes that differences in education are explained by **different costs of education** or by **differences in how signals are perceived**
- **Ability differences** are captured in **benefits (HC)** or in **costs (S)**
- Different policy implications in terms of **efficiency** and **social justice**